

# Winning Space Race With Data Science

## SpaceX Data Analytics

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# Outline

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Executive Summary



Introduction



Methodology



Results



Conclusion



Appendix

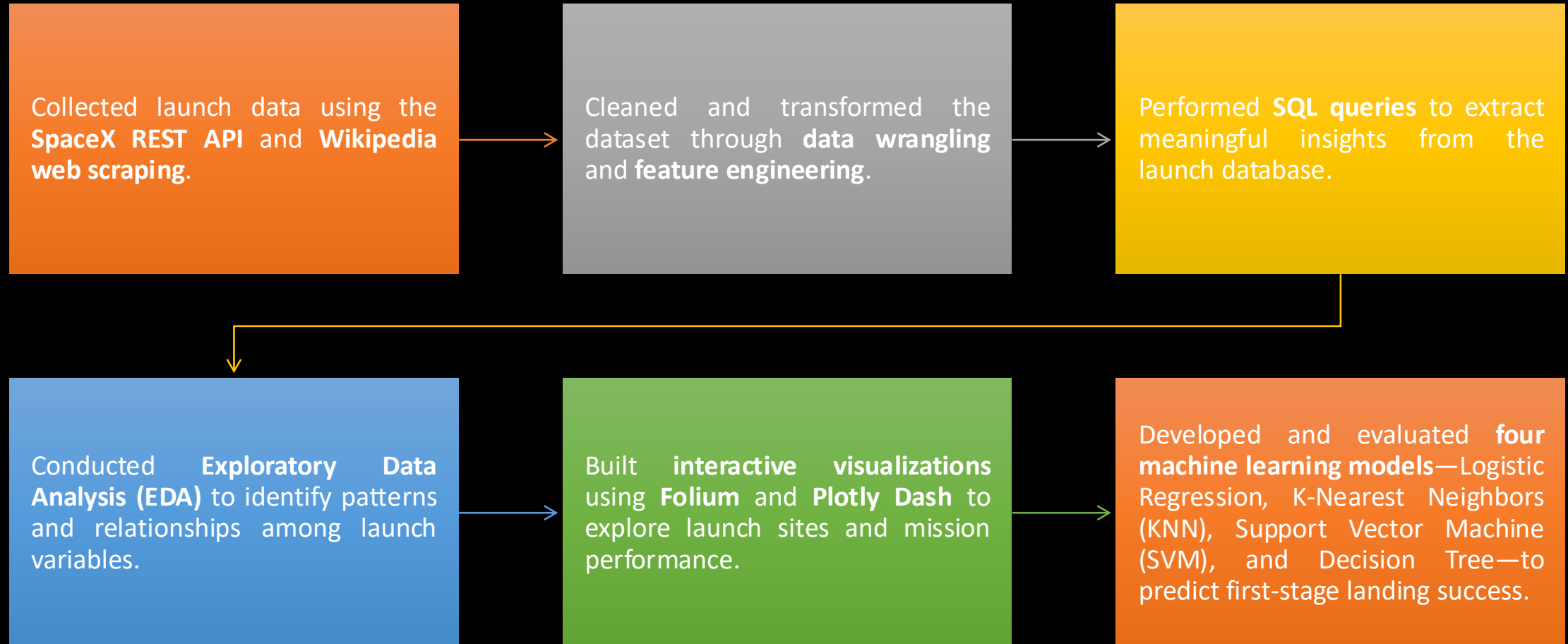
# Section-1

# Executive Summary

## Project Overview

This project analyzes historical SpaceX Falcon 9 launch data to identify the factors influencing first-stage landing success. By leveraging data science techniques, the project aims to predict whether the Falcon 9 first stage can be successfully recovered, helping estimate launch costs and support data-driven decision-making.

# Summary of Methodologies



# Summary of Results



Successfully integrated and analyzed launch data from multiple public data sources.



Identified key factors such as **payload mass, orbit type, launch site, and flight history** that influence landing success.



Created interactive dashboards and geospatial visualizations to communicate launch trends effectively.



All four machine learning models achieved **83.33% prediction accuracy** on the test dataset, demonstrating reliable performance for predicting Falcon 9 first-stage landing outcomes.



The project demonstrates how data science can transform raw aerospace data into actionable insights for launch planning and cost estimation.

# Introduction

## Project Background and Context

The commercial space industry has experienced rapid growth, with companies competing to provide reliable and cost-effective launch services.

**SpaceX** has transformed the industry by developing reusable **Falcon 9** rockets, significantly reducing launch costs through successful first-stage recovery.

Predicting whether the first stage will land successfully is crucial, as reusability directly impacts mission cost, operational efficiency, and launch planning.

This project applies data science and machine learning techniques to analyze historical Falcon 9 launch data and support data-driven decision-making.

# Problem Statement

This project aims to answer the following key questions:

Which launch factors have the greatest influence on Falcon 9 first-stage landing success?

How do payload mass, orbit type, launch site, and flight history affect landing outcomes?

What patterns and trends can be discovered through exploratory data analysis and interactive visualizations?

Can machine learning models accurately predict whether the Falcon 9 first stage will land successfully based on historical launch data?



# Methodology

## Workflow

This project follows a comprehensive data science workflow to analyze historical SpaceX Falcon 9 launch data and predict first-stage landing success.



# Data Collection Methodology

## 1. SpaceX REST API

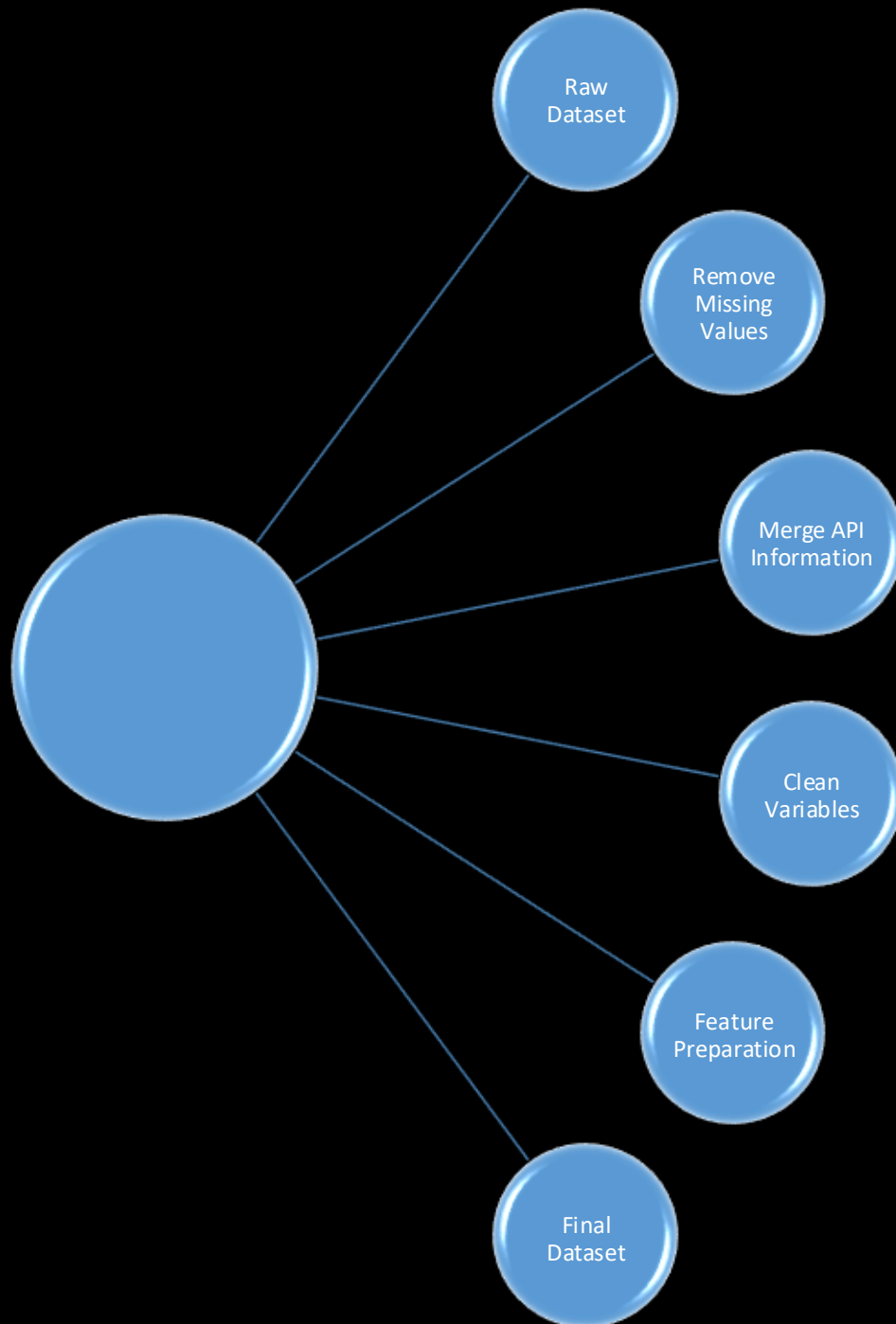
- Retrieved historical Falcon 9 launch records
- Extracted launch details, payload information, boosters, launch pads, and mission outcomes
- Converted JSON responses into structured Pandas DataFrames
- Queried additional API endpoints to enrich the dataset
- [SpaceX data collection via api GitHub file link](#)

# Data Collection Methodology

## 2. Wikipedia Web Scraping

- Scraped Falcon 9 launch records using BeautifulSoup
- Extracted launch dates, payloads, launch sites, customer information, and mission outcomes
- Parsed HTML tables into structured datasets
- Combined scraped information with API data for analysis
- [SpaceX data collection via web scraping GitHub file link](#)

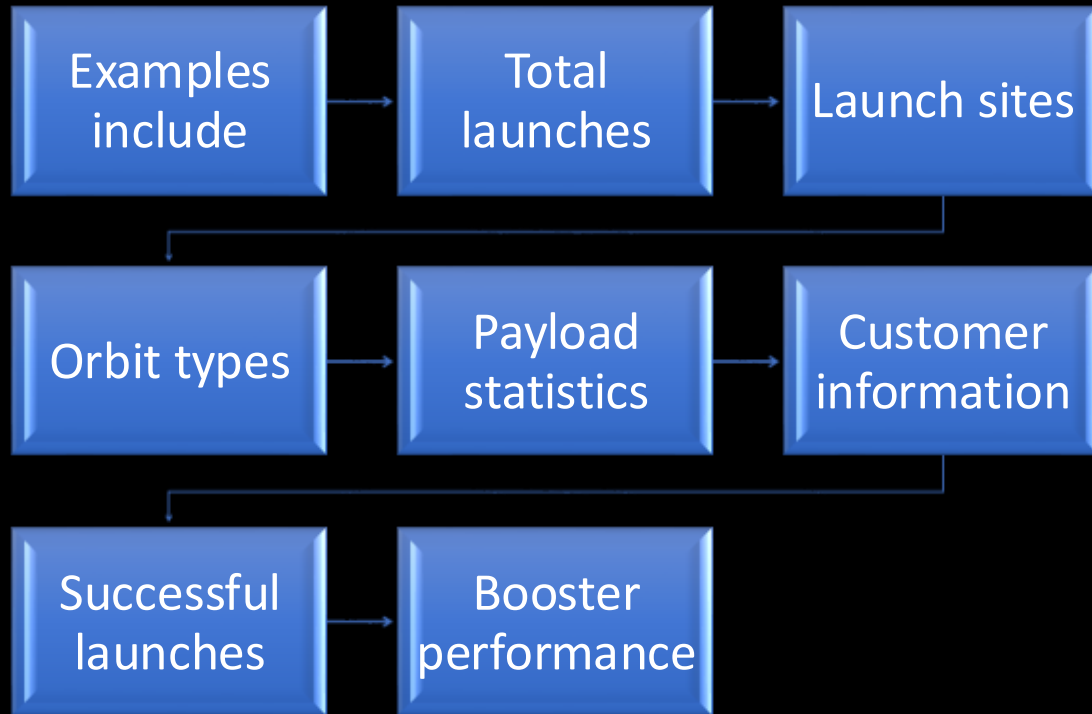
# Data Wrangling



## Data Cleaning Process

- Removed unnecessary columns
- Filtered Falcon 9 launches
- Retrieved additional information using API endpoints
- Handled missing values
- Filled missing Payload Mass values
- Preserved LandingPad NULL values for feature encoding
- Standardized categorical variables
- Generated the final machine learning dataset
- [SpaceX data wrangling GitHub file link](#)

# Exploratory Data Analysis(EDA)



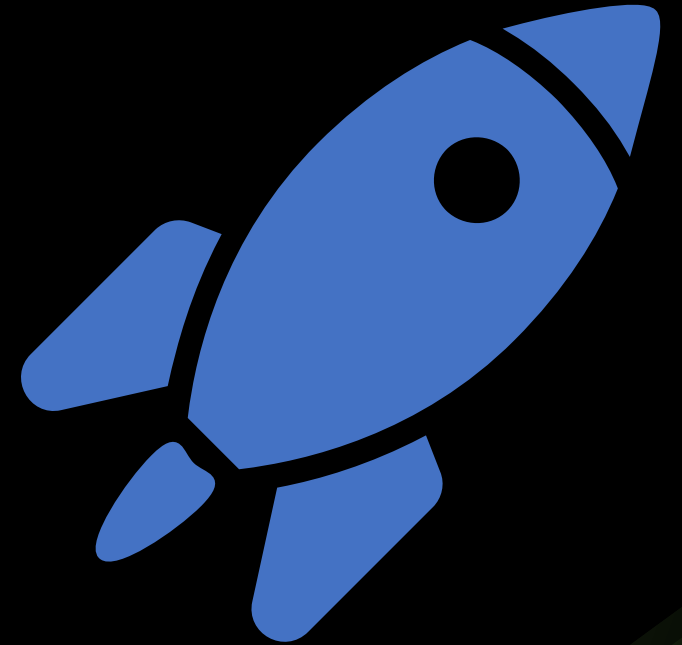
## SQL Analysis:

- Performed SQL queries to explore launch records.
- Ran queries to get answers from the data.

• [SpaceX exploratory data analysis GitHub file link](#)

# Exploratory Data Analysis: Visualization

- Flight Number vs Launch Success
- Payload Mass vs Launch Success
- Orbit vs Launch Success
- Launch Site Performance
- Mission Outcomes
- Booster Version Analysis
- [SpaceX EDA and feature engineering GitHub file link](#)



# Interactive Map using Folium

## Folium Map Components

- Created interactive maps containing
- Launch Site Markers
- Marker Clusters
- Circle Markers
- Distance Lines
- Landing Locations
- Nearby Railways
- Highways
- Coastlines
- Cities

## Purpose

- Compare launch locations
- Analyze geographical factors
- Measure distances to nearby infrastructure
- Visualize successful and failed launches
- [SpaceX data visualization using folium GitHub file link](#)

# Dashboard Development using Plotly Dash

## Dashboard Features

- Developed an interactive dashboard including
- Launch Site Dropdown
- Payload Range Slider
- Success Rate Pie Chart
- Payload vs Success Scatter Plot
- Dynamic User Interaction

## Purpose

- Explore launch performance
- Compare launch sites
- Analyze payload impact
- Support interactive business insights
- [SpaceX data visualization using Plotly Dash GitHub file link](#)

# Predictive Analysis

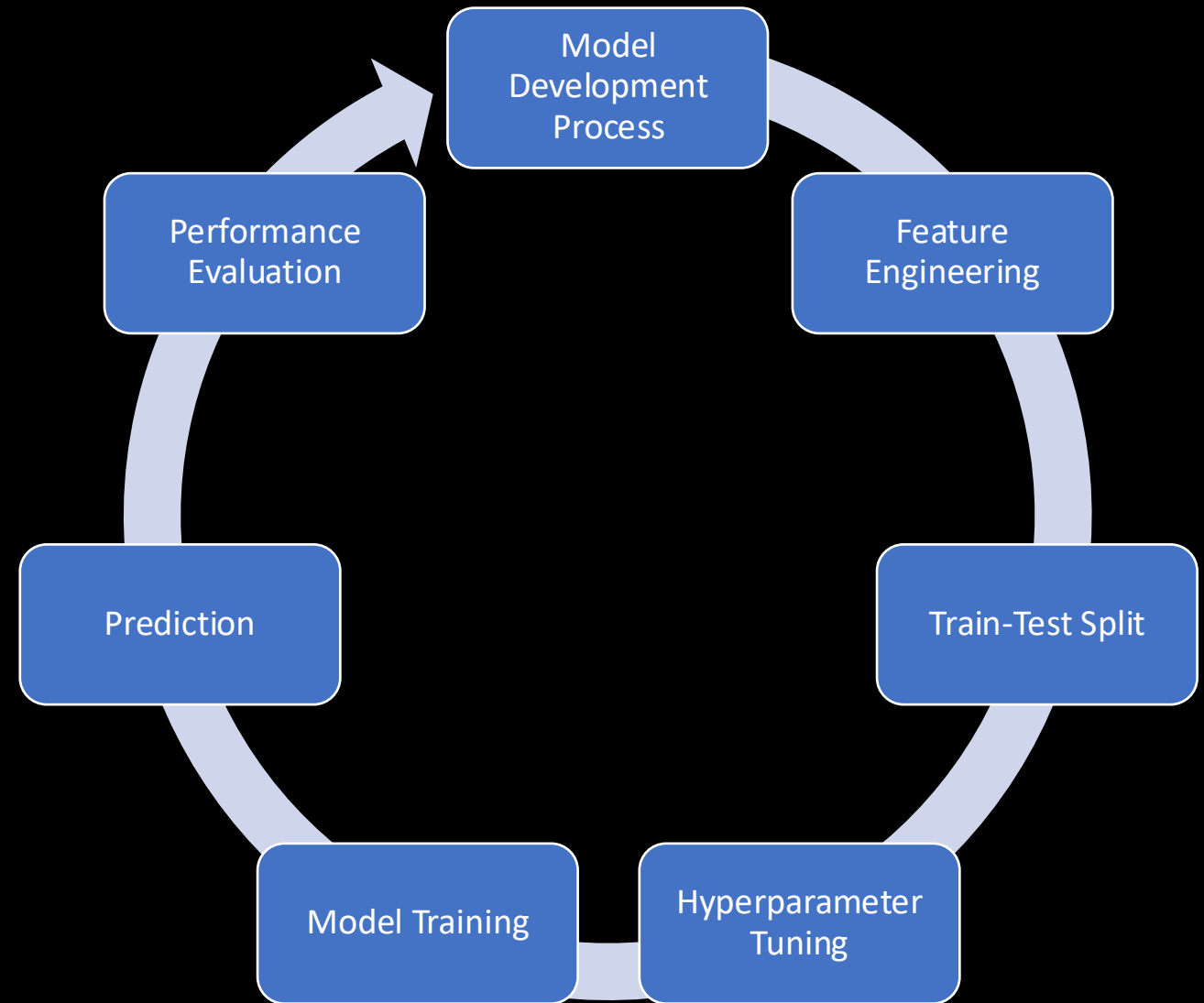
## Classification Models

- Logistic Regression
- K-Nearest Neighbors (KNN)
- Support Vector Machine (SVM)
- Decision Tree

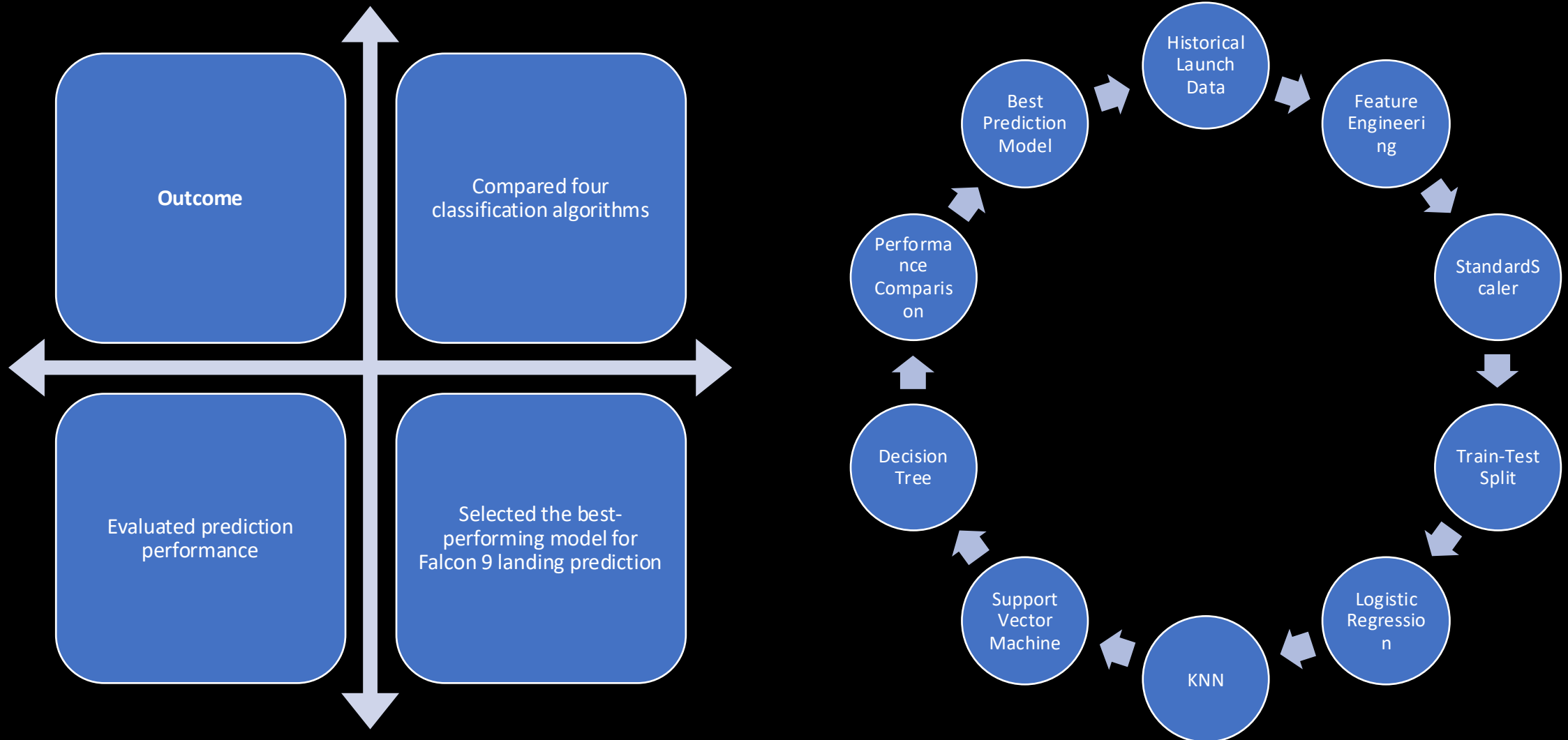
## Evaluation Metrics

- Accuracy
- Confusion Matrix
- Classification Report
- Best Model Comparison

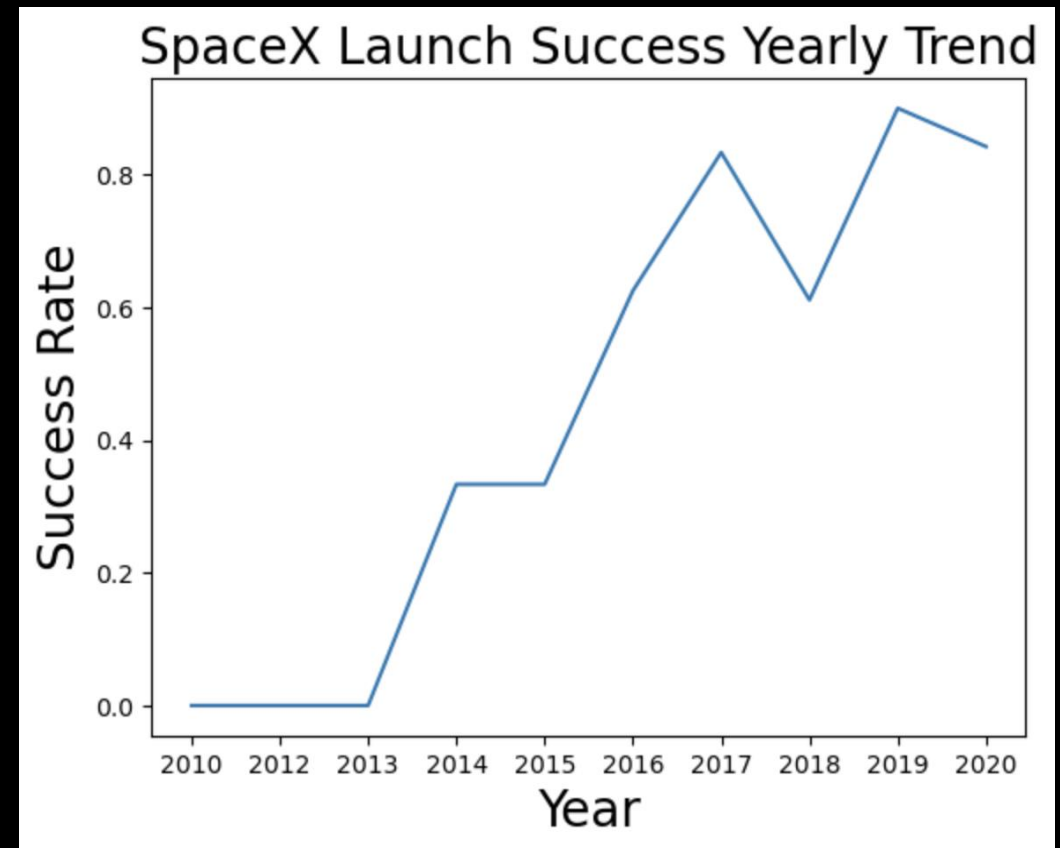
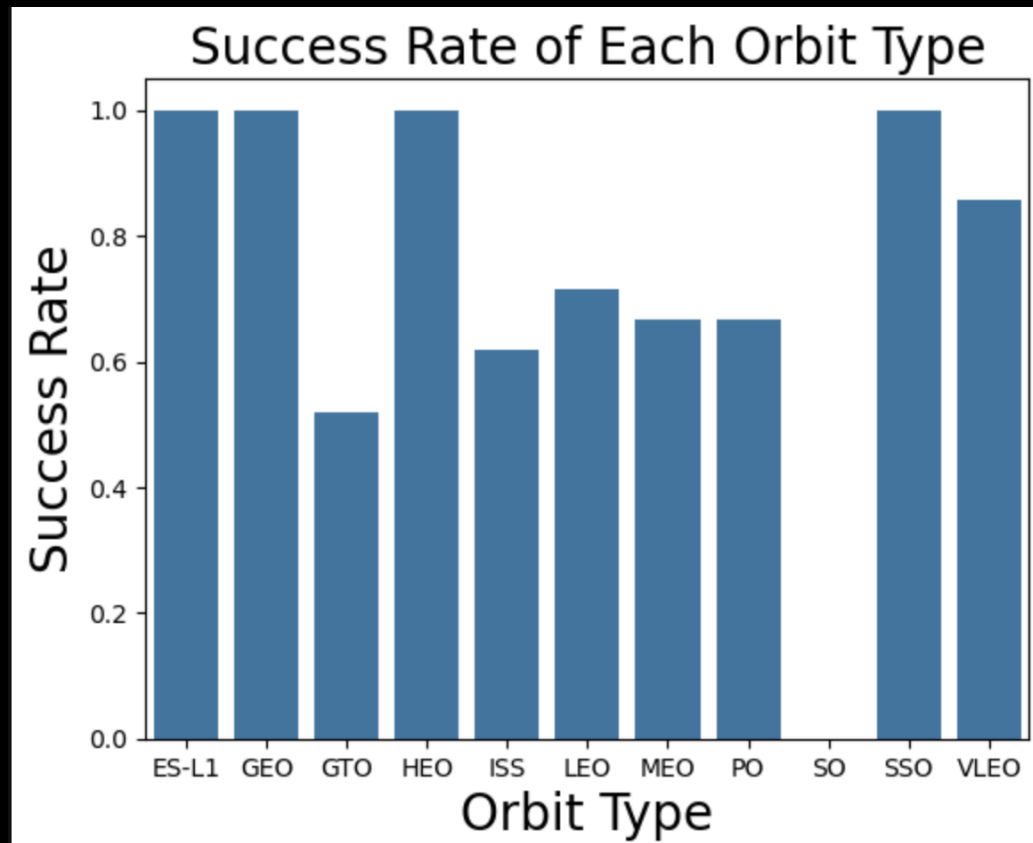
[SpaceX data Machine Learning prediction GitHub file link](#)



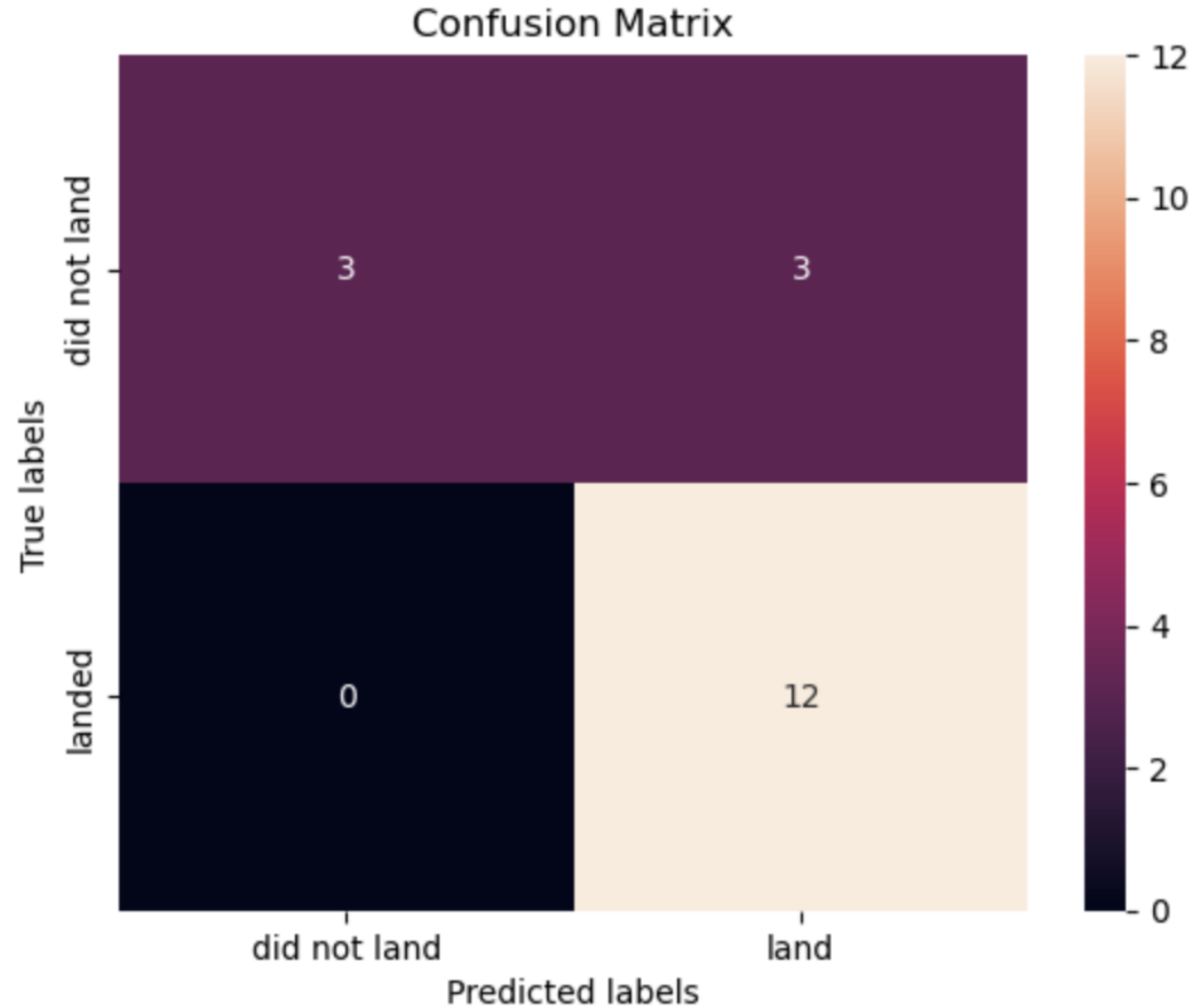
# Machine Learning Workflow



# EDA Results

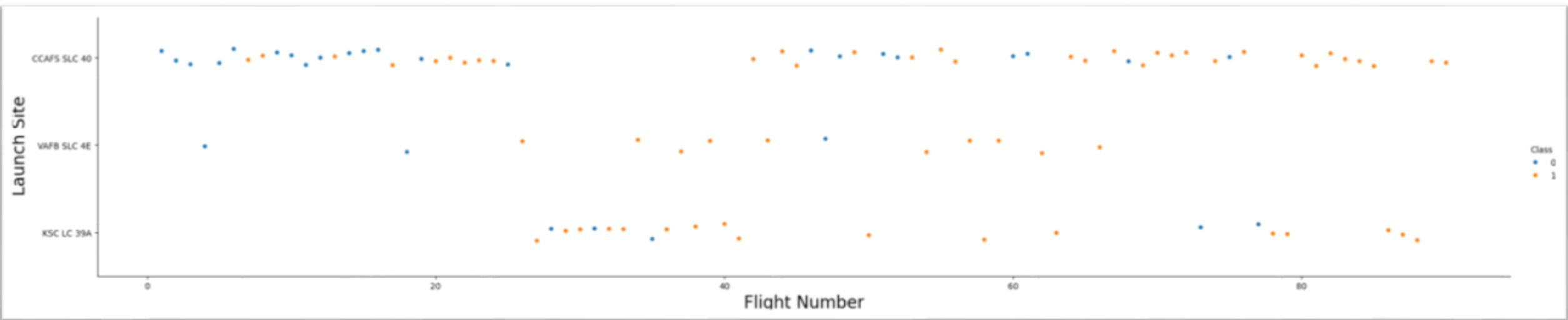


# Predictive Analytics

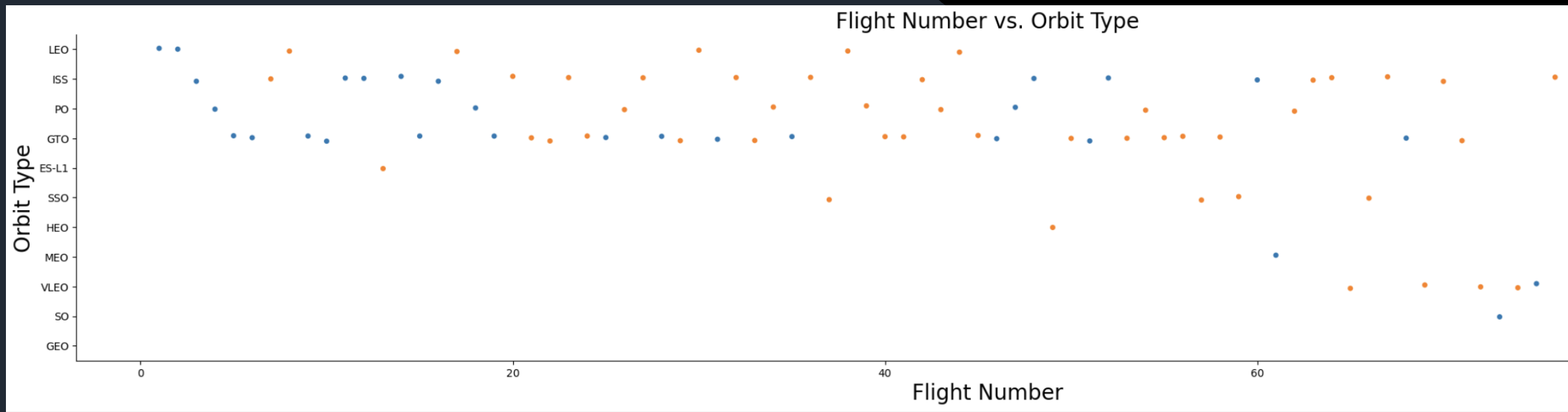


| Model                  | Cross-Validation Accuracy | Test Accuracy |
|------------------------|---------------------------|---------------|
| Decision Tree          | 86.25%                    | 83.33%        |
| Support Vector Machine | 84.82%                    | 83.33%        |
| K-Nearest Neighbors    | 83.39%                    | 83.33%        |
| Logistic Regression    | 82.14%                    | 83.33%        |

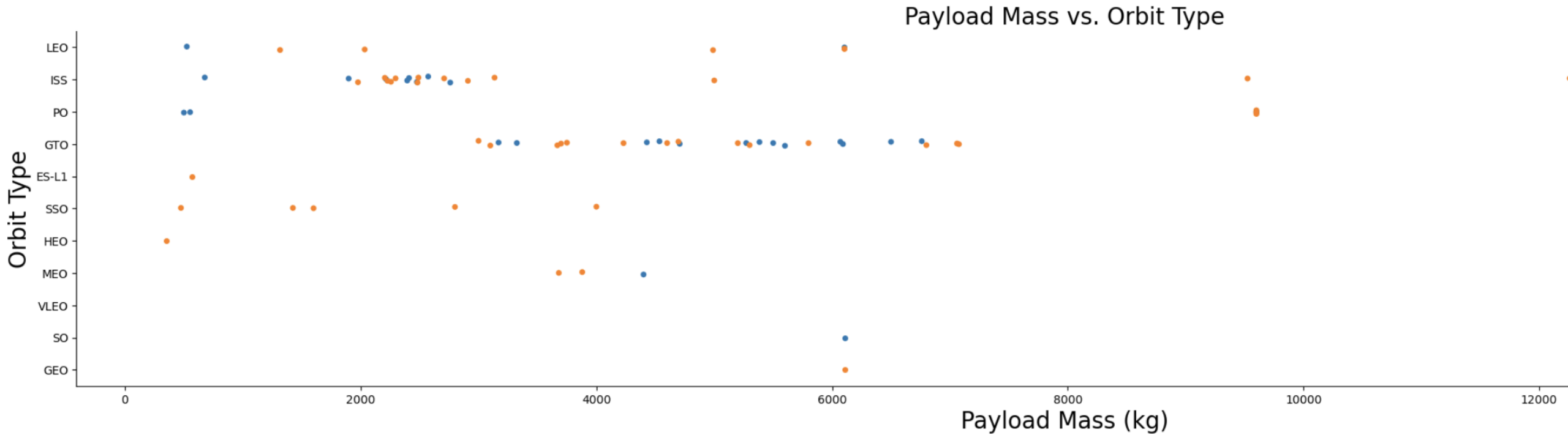
# Section-2



Flight Number vs. Launch Site

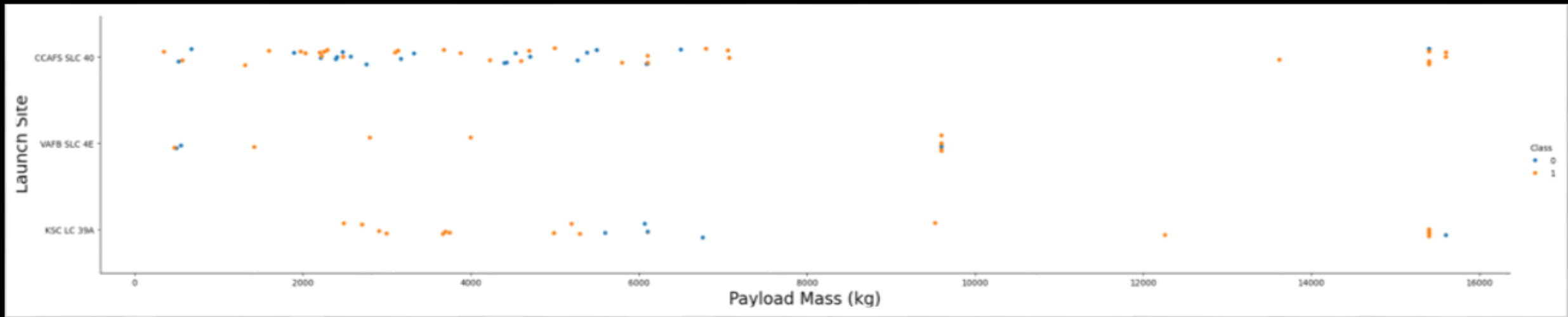


Flight Number vs. Orbit Type

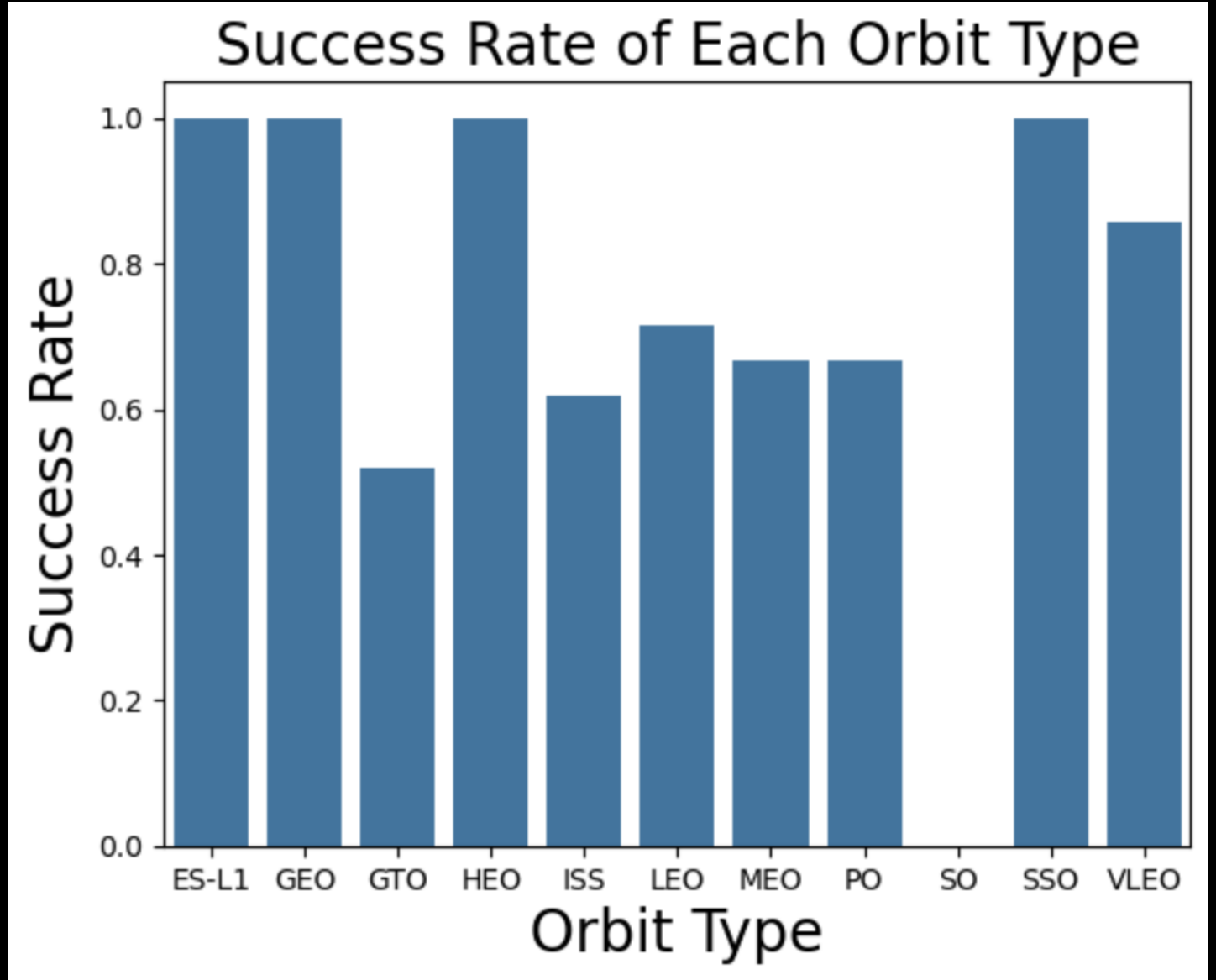


Payload vs.  
Orbit Type

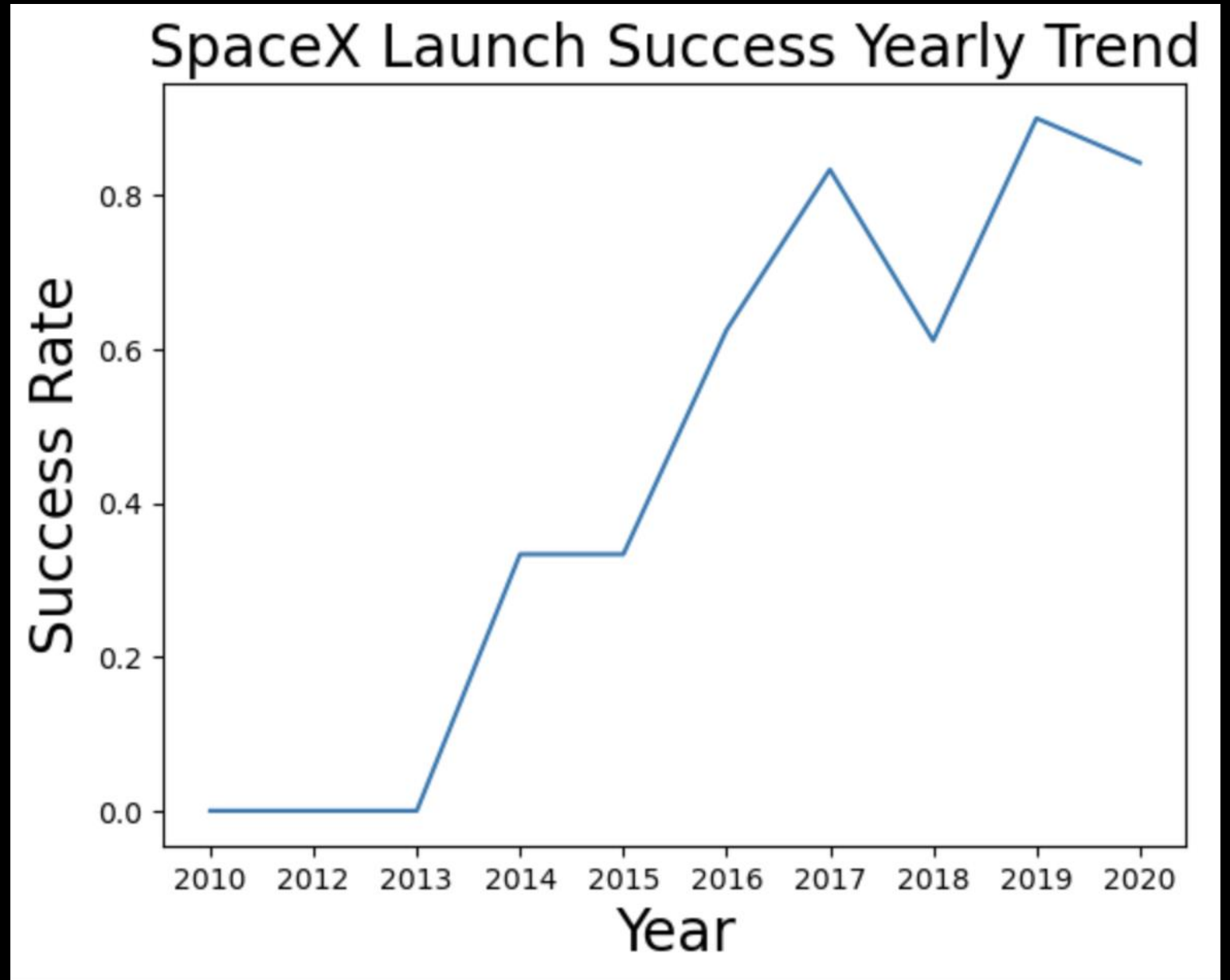
# Payload vs. Launch Site



## Success Rate vs. Orbit Type



# Launch Success Yearly Trend



# SQL Analysis: Unique Launch Site Names

## Explanation

This query retrieves all unique Falcon 9 launch sites from the dataset using the `DISTINCT` keyword. The results show that launches were conducted from four primary launch complexes across Florida and California, providing the basis for comparing launch activity and success rates by location.

### Task 1

Display the names of the unique launch sites in the space mission

```
%sql SELECT DISTINCT Launch_Site FROM SPACEXTABLE;
```

```
* sqlite:///Assets/6_my_data1.db  
Done.
```

| Launch_Site |
|-------------|
|-------------|

|             |
|-------------|
| CCAFS LC-40 |
|-------------|

|             |
|-------------|
| VAFB SLC-4E |
|-------------|

|            |
|------------|
| KSC LC-39A |
|------------|

|              |
|--------------|
| CCAFS SLC-40 |
|--------------|

# SQL Analysis: Launch Sites Beginning with 'CCA'

## Explanation

- This query filters launch records using the LIKE operator to identify launch sites whose names begin with "CCA".
- The LIMIT 5 clause returns the first five matching records, providing a sample of launches conducted from **Cape Canaveral Air Force Station (CCAFS)**.
- The results help analyze launch activities specific to CCAFS and support location-based performance comparisons in later analyses.

## Task 2

Display 5 records where launch sites begin with the string 'CCA'

```
%sql SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

```
* sqlite:///Assets/6_my_data1.db
```

Done.

| Date       | Time (UTC) | Booster_Version | Launch_Site | Payload                                                       | PAYLOAD_MASS__KG_ | Orbit     | Customer        | Mission_Outcome | Landing_Outcome     |
|------------|------------|-----------------|-------------|---------------------------------------------------------------|-------------------|-----------|-----------------|-----------------|---------------------|
| 2010-06-04 | 18:45:00   | F9 v1.0 B0003   | CCAFS LC-40 | Dragon Spacecraft Qualification Unit                          | 0                 | LEO       | SpaceX          | Success         | Failure (parachute) |
| 2010-12-08 | 15:43:00   | F9 v1.0 B0004   | CCAFS LC-40 | Dragon demo flight C1, two CubeSats, barrel of Brouere cheese | 0                 | LEO (ISS) | NASA (COTS) NRO | Success         | Failure (parachute) |
| 2012-05-22 | 7:44:00    | F9 v1.0 B0005   | CCAFS LC-40 | Dragon demo flight C2                                         | 525               | LEO (ISS) | NASA (COTS)     | Success         | No attempt          |
| 2012-10-08 | 0:35:00    | F9 v1.0 B0006   | CCAFS LC-40 | SpaceX CRS-1                                                  | 500               | LEO (ISS) | NASA (CRS)      | Success         | No attempt          |
| 2013-03-01 | 15:10:00   | F9 v1.0 B0007   | CCAFS LC-40 | SpaceX CRS-2                                                  | 677               | LEO (ISS) | NASA (CRS)      | Success         | No attempt          |

# SQL Analysis: Total Payload Mass for NASA Missions

## Explanation

- This query calculates the **total payload mass** delivered by Falcon 9 launches for **NASA Commercial Resupply Services (CRS)** missions.
- The SUM() aggregate function is used to compute the combined payload mass of all qualifying launches.
- The result highlights the significant contribution of Falcon 9 rockets in transporting cargo and supplies for NASA missions, demonstrating SpaceX's role in supporting the **International Space Station (ISS)** and other space operations.

## Task 3

Display the total payload mass carried by boosters launched by NASA (CRS)

```
%sql SELECT SUM(PAYLOAD_MASS_KG_) AS Total_Payload_Mass FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)';
```

```
* sqlite:///Assets/6_my_data1.db  
Done.
```

| Total_Payload_Mass |
|--------------------|
|--------------------|

|       |
|-------|
| 45596 |
|-------|

# SQL Analysis: Average Payload Mass of Falcon 9 v1.1

## Explanation

- This query calculates the **average payload mass** carried by the **Falcon 9 v1.1** booster using the SQL AVG() aggregate function.
- Filtering the dataset by the booster version ensures that only **F9 v1.1** missions are included in the calculation.
- The result provides insight into the typical payload capacity of the Falcon 9 v1.1 rocket and supports comparisons with other booster versions in terms of launch performance.

## Task 4

Display average payload mass carried by booster version F9 v1.1

```
%sql SELECT AVG(PAYLOAD_MASS__KG_) AS Average_Payload_Mass FROM SPACEXTABLE WHERE Booster_Version = 'F9 v1.1';
```

```
* sqlite:///Assets/6_my_data1.db
```

Done.

| Average_Payload_Mass |
|----------------------|
|----------------------|

|        |
|--------|
| 2928.4 |
|--------|

# SQL Analysis: First Successful Ground Landing Date

## Explanation

- This query identifies the **earliest successful Falcon 9 ground pad landing** using the SQL MIN() aggregate function.
- It filters the dataset to include only missions with a **"Success (ground pad)"** landing outcome.
- The result marks a significant milestone in SpaceX's reusable rocket program, demonstrating the beginning of successful first-stage recovery on a ground landing pad.

## Task 5

List the date when the first succesful landing outcome in ground pad was acheived.

*Hint: Use min function*

```
%sql SELECT MIN(Date) AS First_Success_Ground_Pad FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)';
```

```
* sqlite:///Assets/6_my_data1.db
```

```
Done.
```

```
First_Success_Ground_Pad
```

```
2015-12-22
```

# SQL Analysis: Successful Drone Ship Landings (Payload 4,000–6,000 kg)

## Explanation

- This query identifies **Falcon 9 booster versions** that successfully landed on a **drone ship** while carrying payloads within the **4,000–6,000 kg** range.
- Multiple filtering conditions are applied using the WHERE clause to restrict the results by **landing outcome** and **payload mass**.
- The results demonstrate the capability of Falcon 9 boosters to safely recover after delivering medium-weight payloads, highlighting the effectiveness of SpaceX's reusable launch system.

## Task 6

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
%sql SELECT Booster_Version FROM SPACEXTABLE WHERE  
      Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS_KG_ > 4000 AND PAYLOAD_MASS_KG_ < 6000;
```

```
* sqlite:///Assets/6_my_data1.db  
Done.
```

| Booster_Version |
|-----------------|
|-----------------|

|             |
|-------------|
| F9 FT B1022 |
|-------------|

|             |
|-------------|
| F9 FT B1026 |
|-------------|

|               |
|---------------|
| F9 FT B1021.2 |
|---------------|

|               |
|---------------|
| F9 FT B1031.2 |
|---------------|

# SQL Analysis: Total Successful and Failed Mission Outcomes

## Task 7

List the total number of successful and failure mission outcomes ¶

```
%sql SELECT Mission_Outcome, COUNT(*) AS Total_Count FROM SPACEXTABLE GROUP BY Mission_Outcome;
```

```
* sqlite:///Assets/6_my_data1.db  
Done.
```

| Mission_Outcome                  | Total_Count |
|----------------------------------|-------------|
| Failure (in flight)              | 1           |
| Success                          | 98          |
| Success                          | 1           |
| Success (payload status unclear) | 1           |

## Explanation

- This query groups all launch missions by their **mission outcome** and counts the total number of successful and failed missions using the SQL COUNT() function.
- The GROUP BY clause categorizes launches into success and failure, providing a summary of overall mission performance.
- The results indicate the reliability of Falcon 9 launches and establish a baseline for comparing mission success throughout the project.

# SQL Analysis: Booster(s) Carrying the Maximum Payload

## Explanation

- This query identifies the **Falcon 9 booster version(s)** that carried the **highest payload mass** recorded in the dataset.
- The SQL MAX() aggregate function is used to determine the maximum payload value, while a subquery retrieves the corresponding booster version(s).
- The results highlight the booster configuration capable of supporting the heaviest payload missions, demonstrating Falcon 9's evolving launch capacity.

## Task 8

List all the booster\_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.

```
%sql SELECT Booster_Version, PAYLOAD_MASS_KG_ FROM SPACEXTABLE WHERE  
PAYLOAD_MASS_KG_ = (SELECT MAX(PAYLOAD_MASS_KG_) FROM SPACEXTABLE);
```

```
* sqlite:///Assets/6_my_data1.db  
Done.
```

| Booster_Version | PAYLOAD_MASS_KG_ |
|-----------------|------------------|
| F9 B5 B1048.4   | 15600            |
| F9 B5 B1049.4   | 15600            |
| F9 B5 B1051.3   | 15600            |
| F9 B5 B1056.4   | 15600            |
| F9 B5 B1048.5   | 15600            |
| F9 B5 B1051.4   | 15600            |
| F9 B5 B1049.5   | 15600            |
| F9 B5 B1060.2   | 15600            |
| F9 B5 B1058.3   | 15600            |
| F9 B5 B1051.6   | 15600            |
| F9 B5 B1060.3   | 15600            |
| F9 B5 B1049.7   | 15600            |

## Task 9

List the records which will display the month names, failure landing\_outcomes in drone ship ,booster versions, launch\_site for the months in year 2015.

**Note: SQLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.**

```
%sql SELECT substr(Date, 6, 2) AS Month, Landing_Outcome, Booster_Version, Launch_Site FROM SPACEXTABLE
WHERE Landing_Outcome = 'Failure (drone ship)' AND substr(Date, 1, 4) = '2015';
```

```
* sqlite:///Assets/6_my_data1.db
Done.
```

| Month | Landing_Outcome      | Booster_Version | Launch_Site |
|-------|----------------------|-----------------|-------------|
| 01    | Failure (drone ship) | F9 v1.1 B1012   | CCAFS LC-40 |
| 04    | Failure (drone ship) | F9 v1.1 B1015   | CCAFS LC-40 |

# SQL Analysis: Failed Drone Ship Landings in 2015

## Explanation

- This query retrieves all **2015 Falcon 9 missions** that experienced **failed drone ship landings**.
- It filters the dataset by **launch year (2015)** and **landing outcome**, returning the corresponding **booster versions** and **launch site names**.
- The results provide insight into the early challenges of SpaceX's reusable rocket program before drone ship landing technology became consistently successful.

## Task 10

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

```
%sql SELECT Landing_Outcome, COUNT(*) AS Outcome_Count FROM SPACEXTABLE WHERE Date  
|BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY Outcome_Count DESC;
```

```
* sqlite:///Assets/6_my_data1.db  
Done.
```

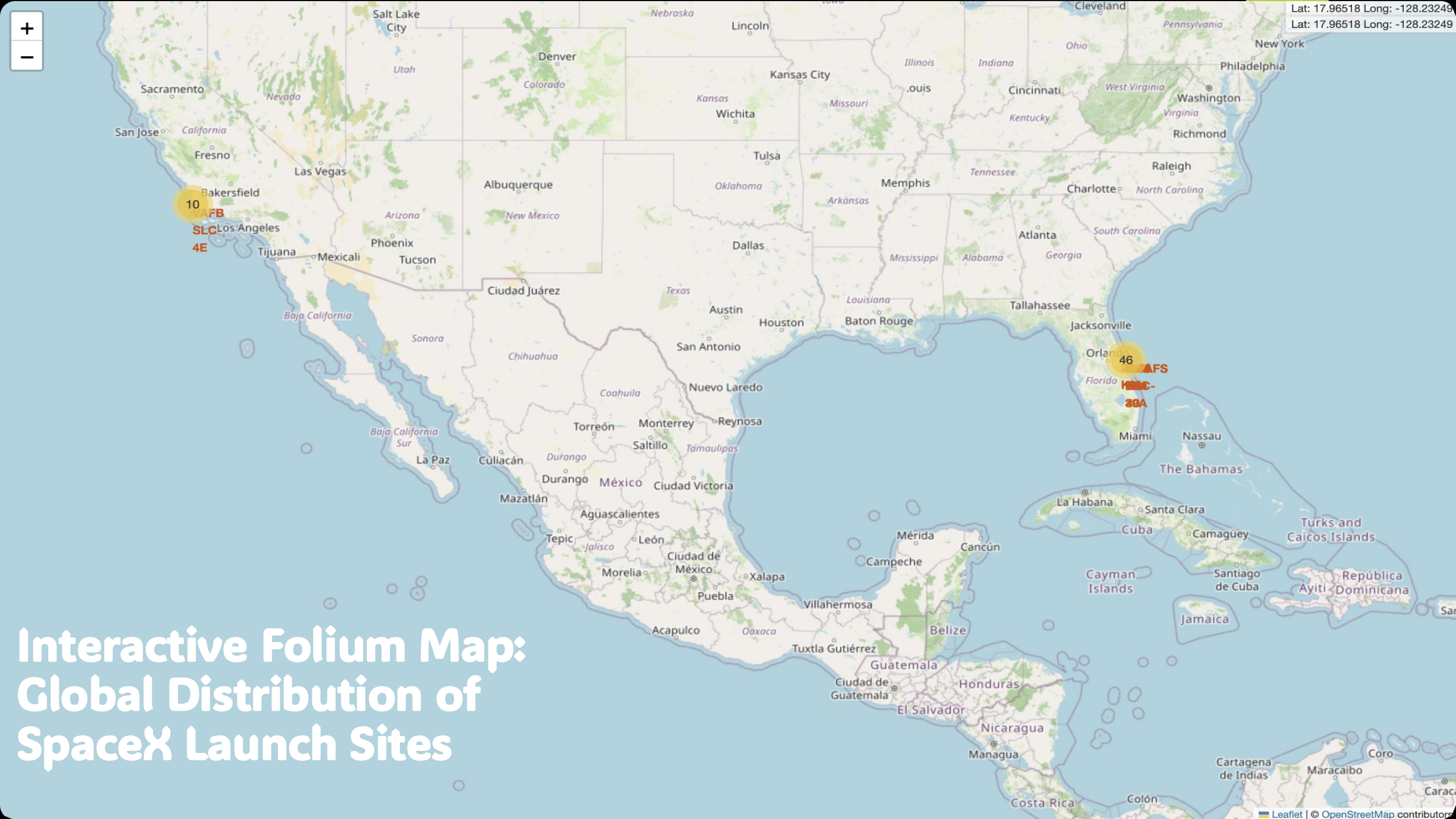
| Landing_Outcome        | Outcome_Count |
|------------------------|---------------|
| No attempt             | 10            |
| Success (drone ship)   | 5             |
| Failure (drone ship)   | 5             |
| Success (ground pad)   | 3             |
| Controlled (ocean)     | 3             |
| Uncontrolled (ocean)   | 2             |
| Failure (parachute)    | 2             |
| Precluded (drone ship) | 1             |

# SQL Analysis: Ranking Landing Outcomes (2010–2017)

## Explanation

- This query counts the occurrence of each **landing outcome** for Falcon 9 missions between **June 4, 2010**, and **March 20, 2017**.
- The results are grouped by **landing outcome** and ranked in **descending order** by frequency.
- The ranking provides an overview of landing performance during the early years of Falcon 9 operations, highlighting the most common recovery outcomes and supporting trend analysis.

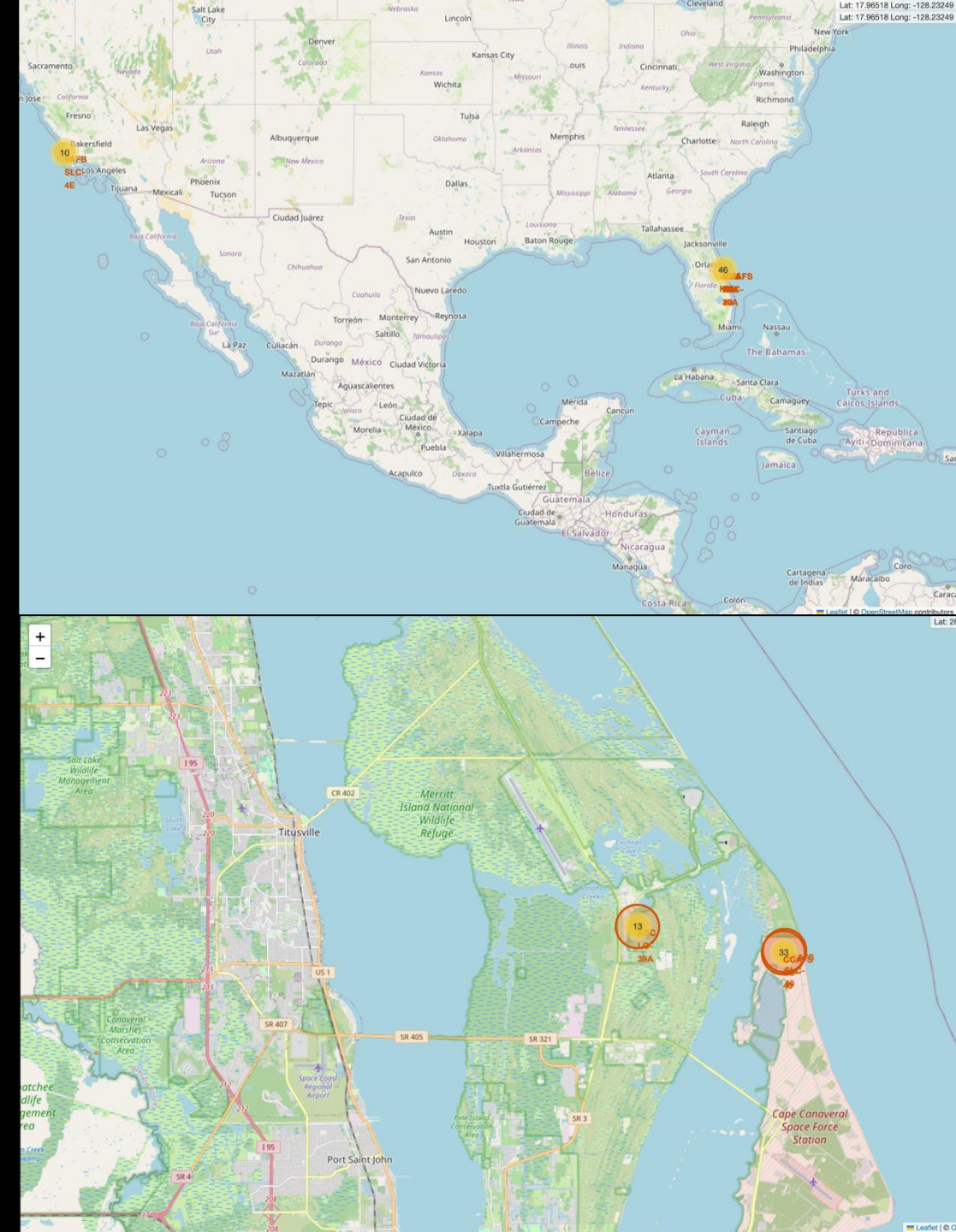
# Section-3

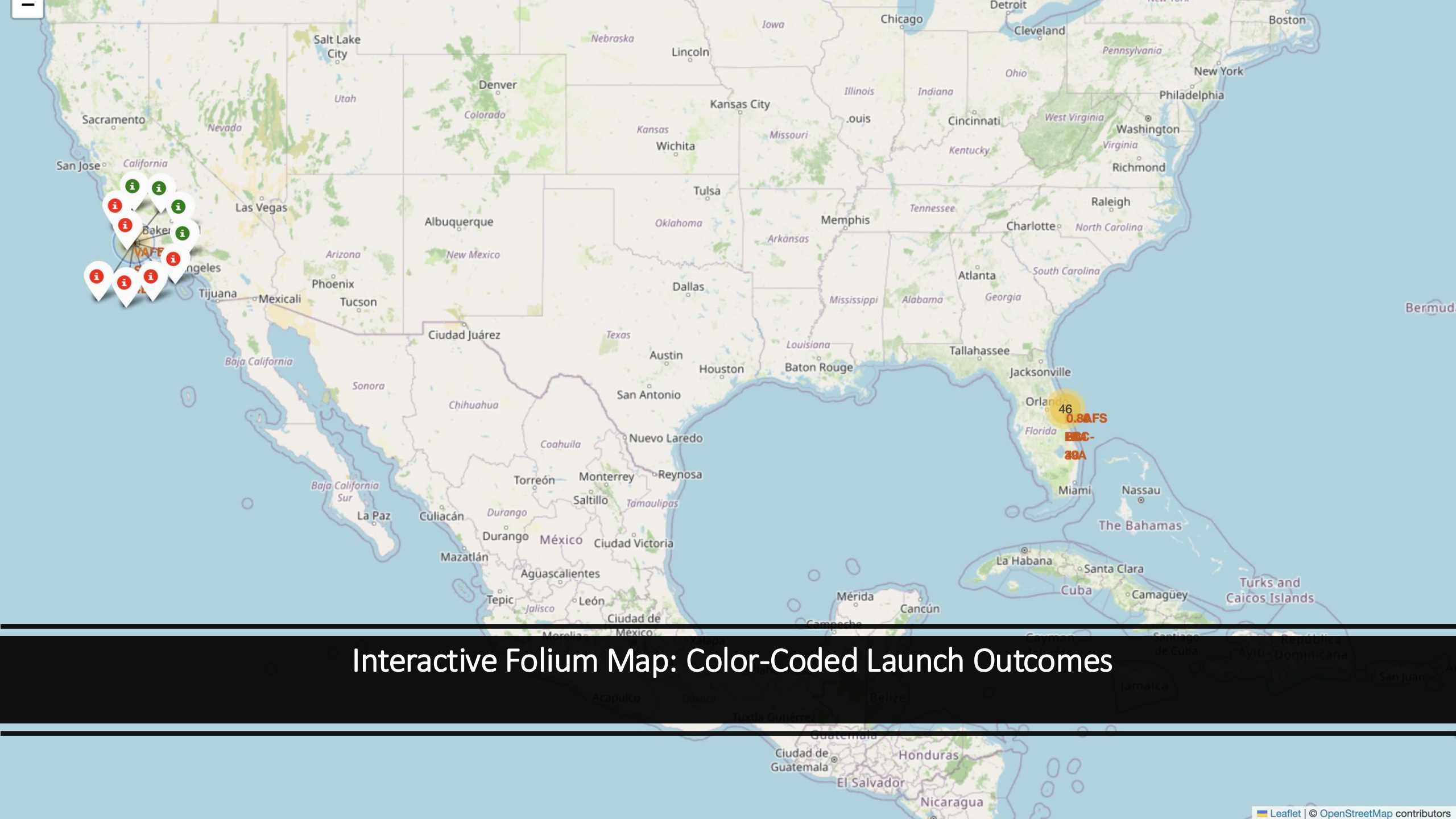


# Interactive Folium Map: Global Distribution of SpaceX Launch Sites

## Map Elements

- **Location Markers:** Indicate the geographic coordinates of all SpaceX Falcon 9 launch sites.
- **Interactive Pop-ups:** Display launch site names and related launch information when a marker is selected.
- **Global Basemap:** Provides geographical context by showing the relative positions of launch sites.
- **Zoom & Navigation Controls:** Allow users to interactively explore different regions of the map.

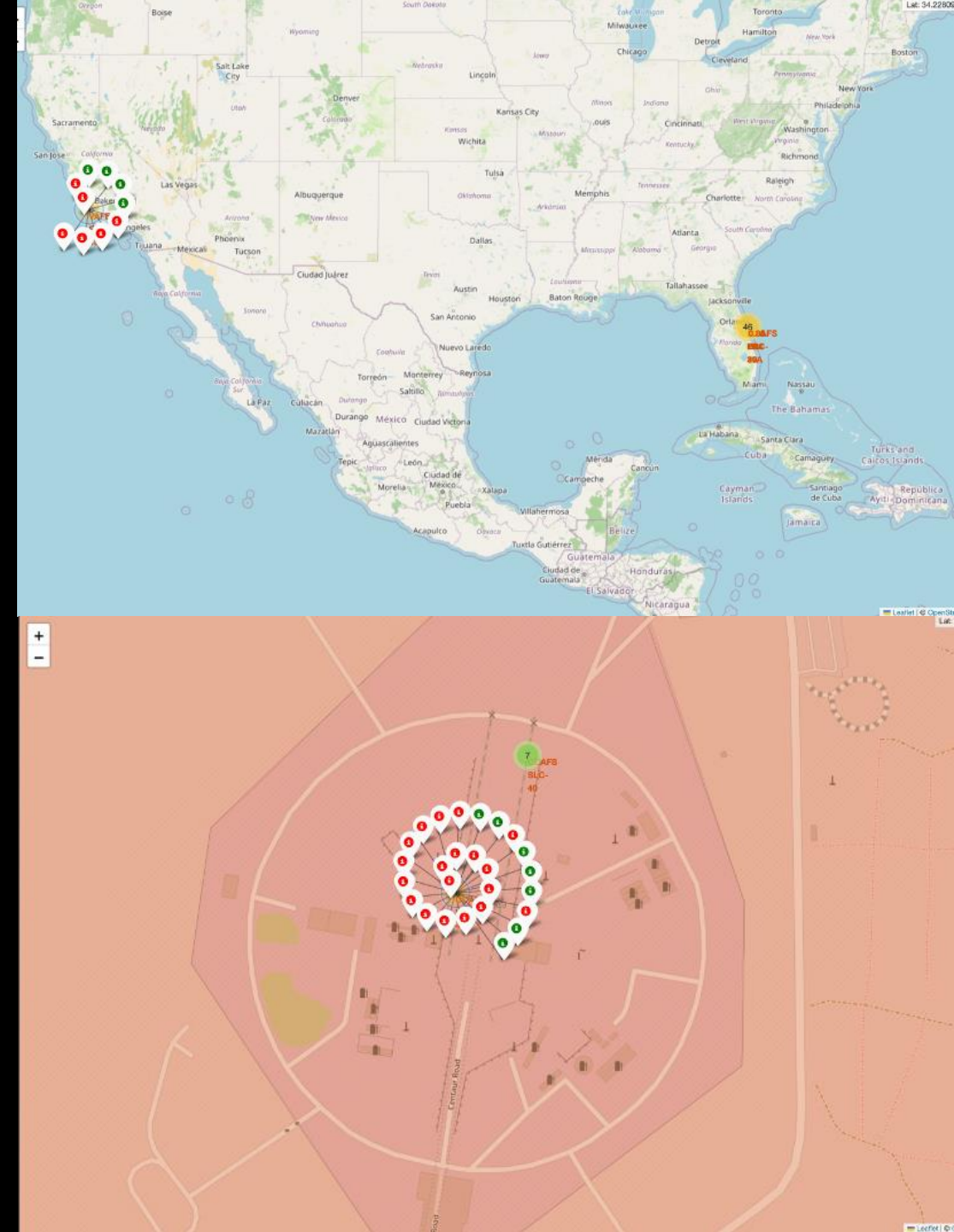


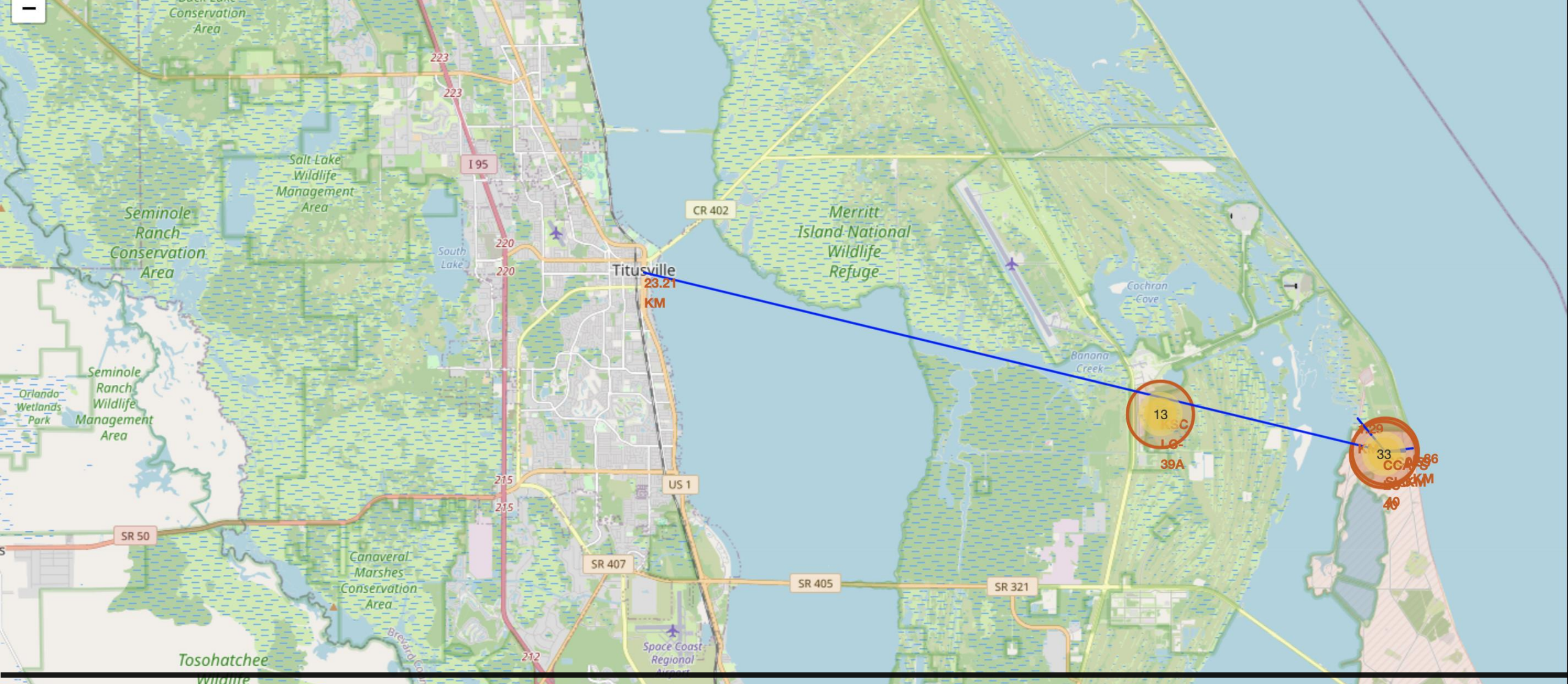


# Interactive Folium Map: Color-Coded Launch Outcomes

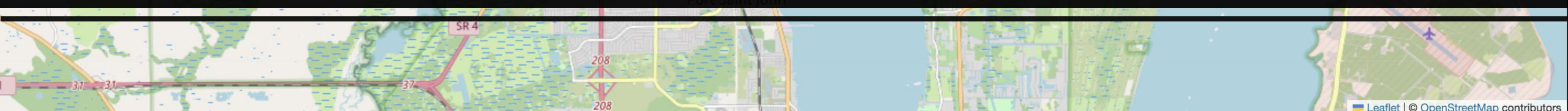
## Map Elements

- **Green Markers:** Represent launches with successful first-stage landings.
- **Red Markers:** Represent launches with unsuccessful landing attempts.
- **Launch Site Markers:** Display the geographic locations of Falcon 9 launch sites.
- **Interactive Pop-ups:** Provide mission details when a marker is selected.





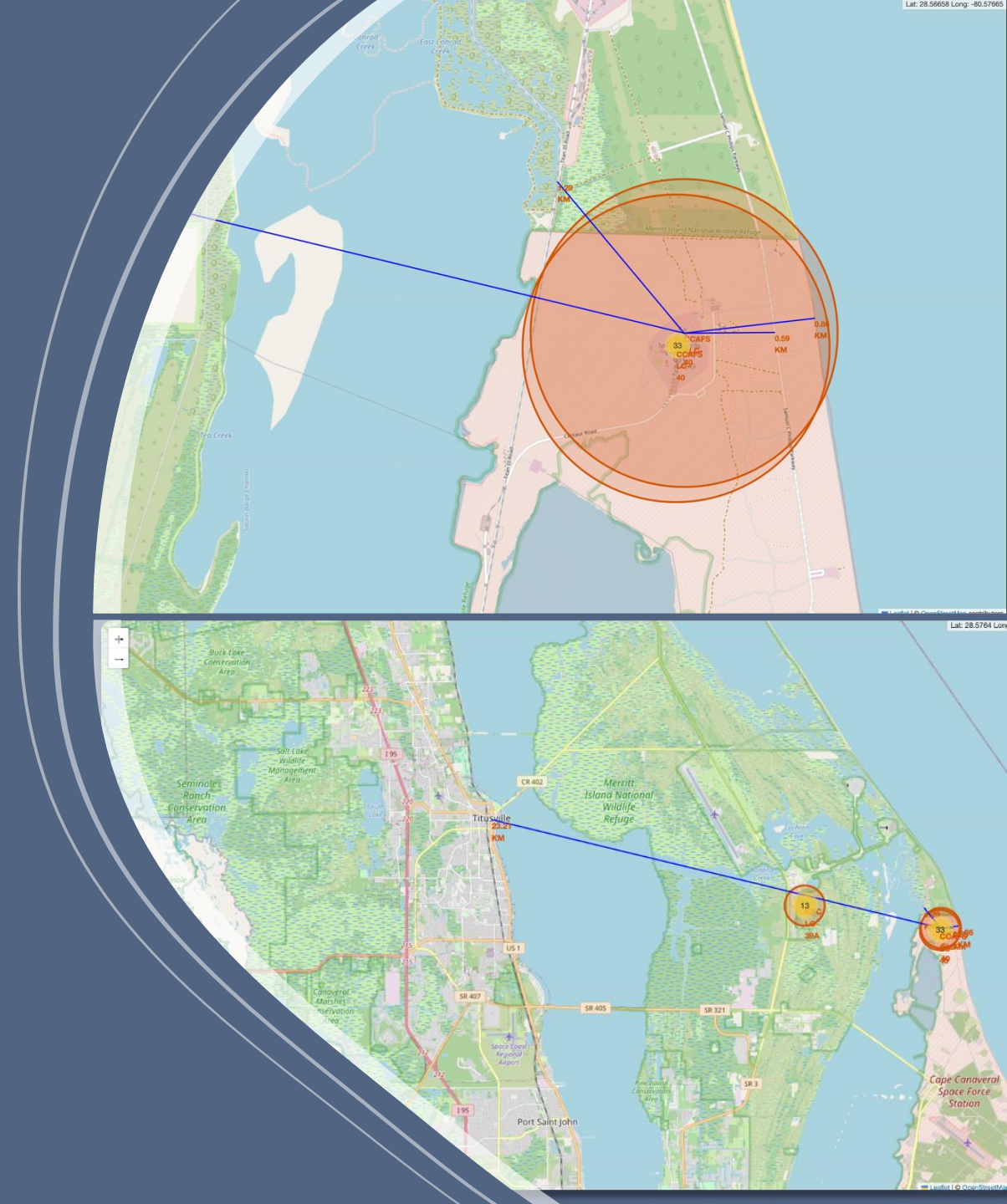
# Interactive Folium Map: Launch Site Proximity Analysis



# Interactive Folium Map: Launch Site Proximity Analysis

## Map Elements

- **Launch Site Marker:** Indicates the selected Falcon 9 launch complex.
- **Distance Lines:** Connect the launch site to nearby infrastructure, such as railways, highways, and the coastline.
- **Distance Labels:** Display the calculated distances between the launch site and surrounding features.
- **Infrastructure Markers:** Highlight nearby transportation routes and geographical landmarks.



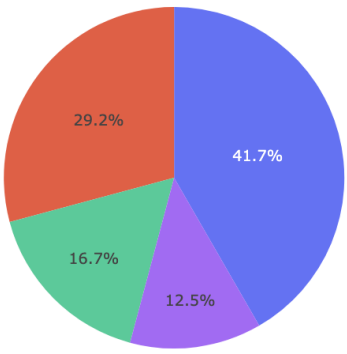
# Section-4

# Interactive Dashboard: Launch Success Distribution Across All Sites

All Sites



Total Successful Launches by Site

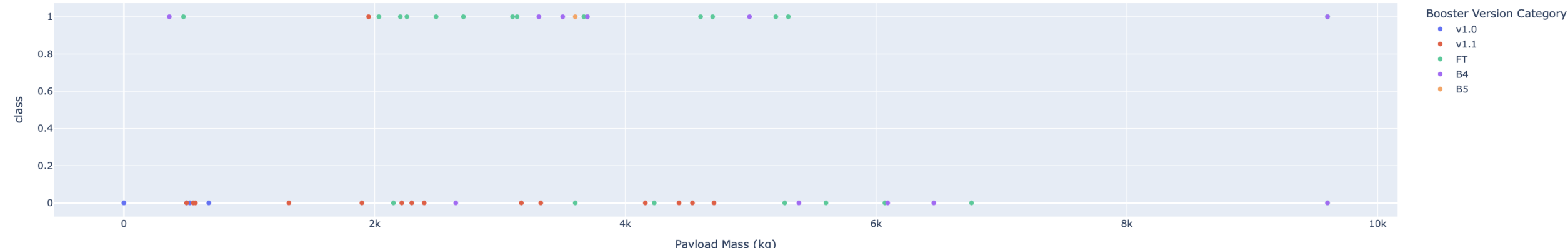


- KSC LC-39A
- CCAFS LC-40
- VAFB SLC-4E
- CCAFS SLC-40

Payload range (Kg):



Payload vs Launch Outcome for All Sites



# Interactive Dashboard: Launch Success Distribution Across All Sites



## Dashboard Elements



**Pie Chart:** Visualizes the distribution of successful launches across all SpaceX launch sites.



**Launch Site Selector:** Configured to display aggregated results for **all sites**.



**Interactive Dashboard:** Enables users to explore launch performance dynamically.

# Interactive Dashboard: Launch Success Distribution Across All Sites

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## Key Findings



The pie chart provides a comparative view of successful launches across all launch sites.



It highlights how launch activity and success are distributed among different SpaceX launch complexes.



The visualization enables quick identification of launch sites with the highest contribution to successful missions.



Interactive filtering allows users to compare launch performance without manually analyzing the dataset.

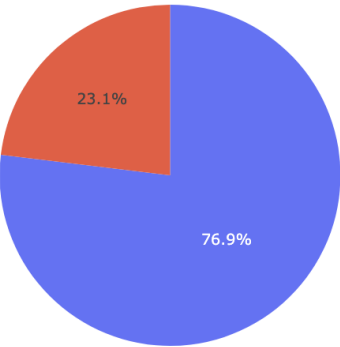
# Interactive Dashboard: Launch Success Ratio at the Best-Performing Launch Site

KSC LC-39A

✕ ▾



Success vs Failure for KSC LC-39A



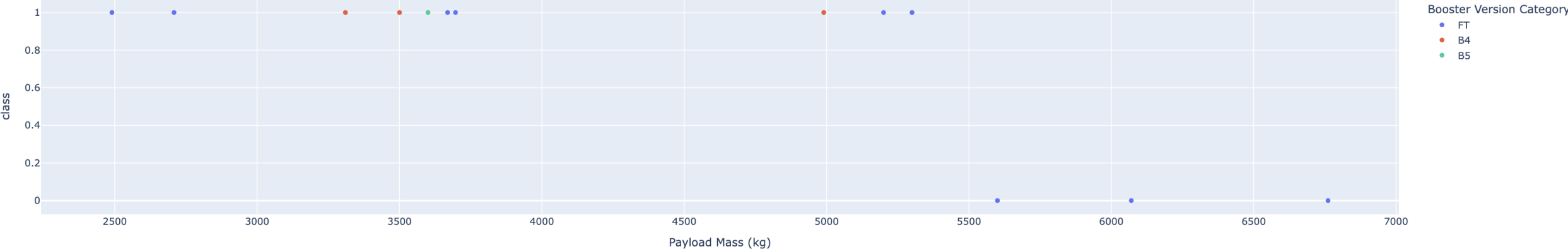
Success

Failure

Payload range (Kg):



Payload vs Launch Outcome for KSC LC-39A



# Interactive Dashboard: Launch Success Ratio at the Best-Performing Launch Site

## Dashboard Elements

- **Filtered Pie Chart:** Displays successful and failed launches for a selected launch site.
- **Launch Site Dropdown:** Allows users to focus on individual launch sites.
- **Interactive Visualization:** Updates automatically based on the selected site.



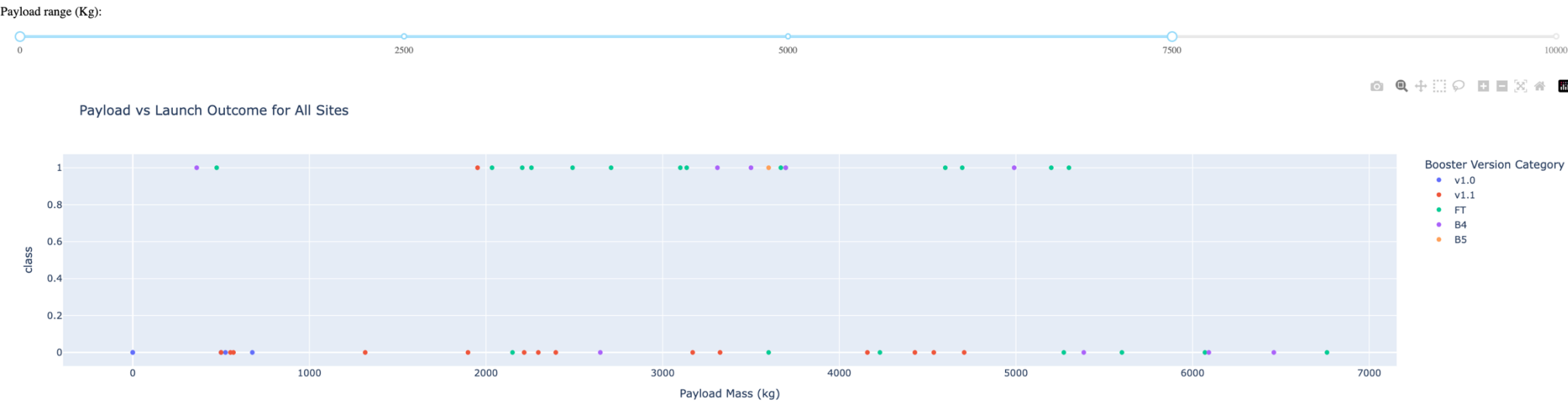
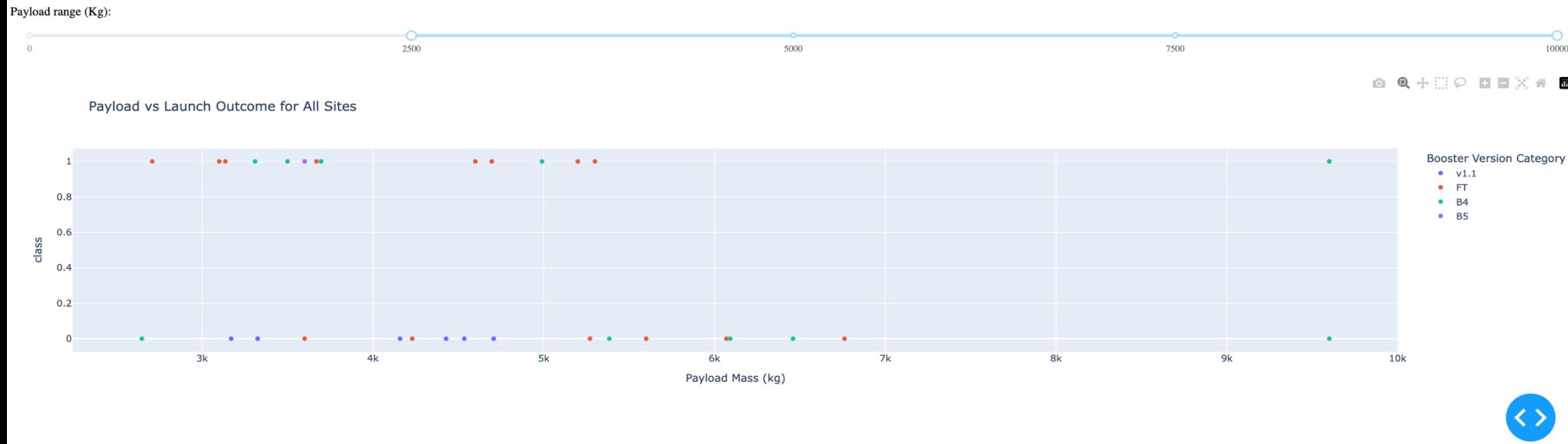
# Interactive Dashboard: Launch Success Ratio at the Best-Performing Launch Site

## Key Findings

- The pie chart illustrates the success-to-failure ratio for the selected launch site.
- The selected site demonstrates the **highest launch success ratio**, indicating consistent mission performance.
- Comparing individual launch sites helps evaluate operational reliability and identify top-performing launch facilities.
- Interactive filtering provides detailed, site-specific insights that are not visible in the aggregated view.



# Interactive Dashboard: Payload Mass vs. Launch Success Analysis



# Interactive Dashboard: Payload Mass vs. Launch Success Analysis



## Dashboard Elements



**Scatter Plot:** Displays the relationship between payload mass and launch outcome.



**Payload Range Slider:** Filters missions based on payload mass.

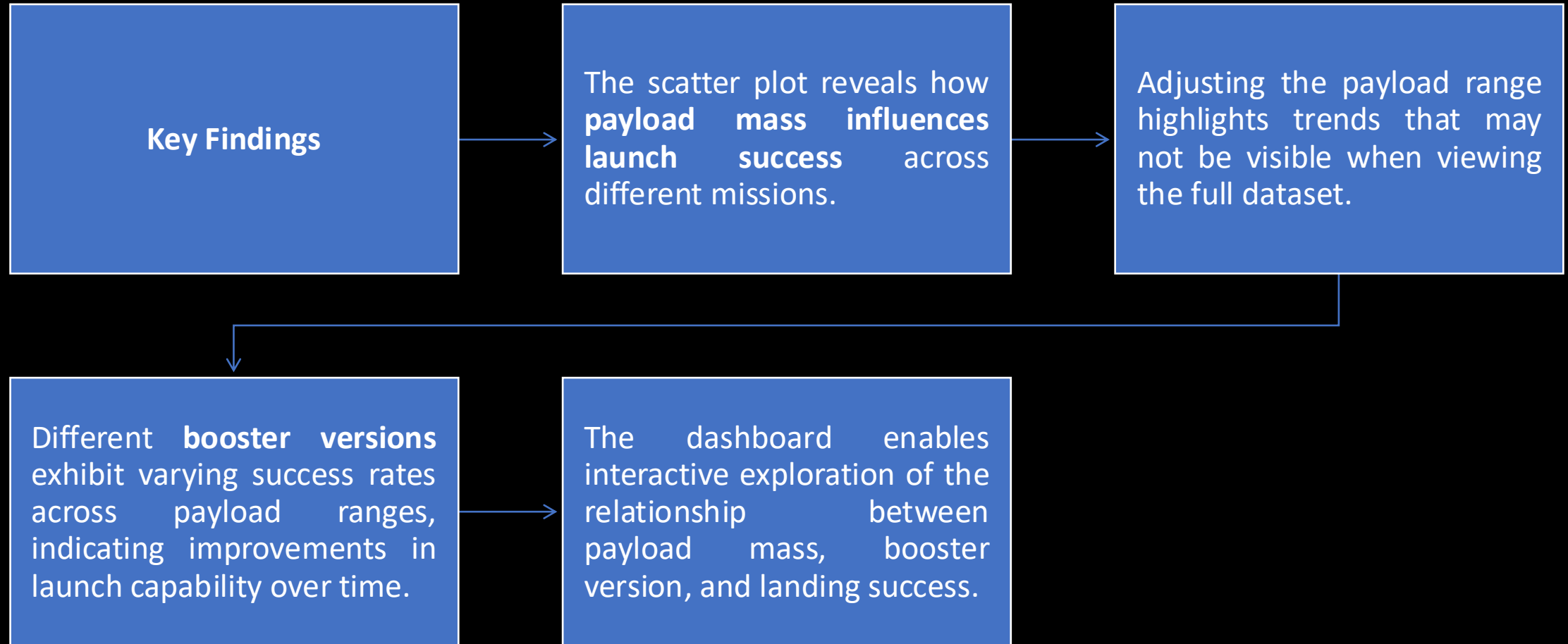


**Booster Version Colours:** Different colours represent different Falcon 9 booster versions.



**Interactive Controls:** Allow users to dynamically explore launch performance.

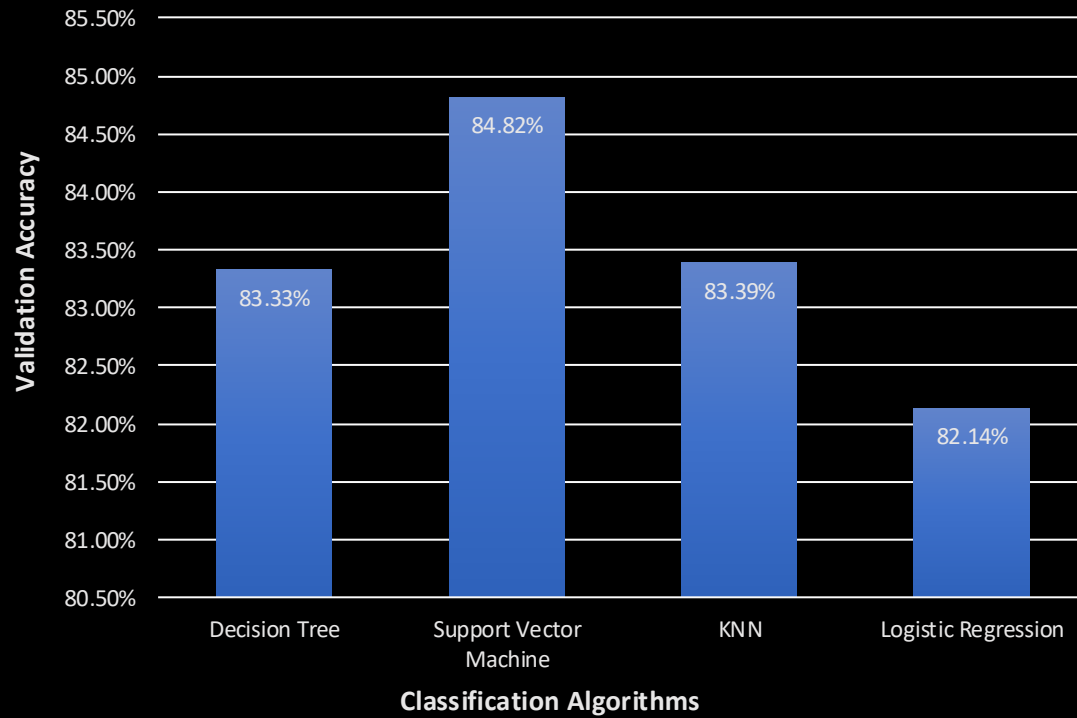
# Interactive Dashboard: Payload Mass vs. Launch Success Analysis



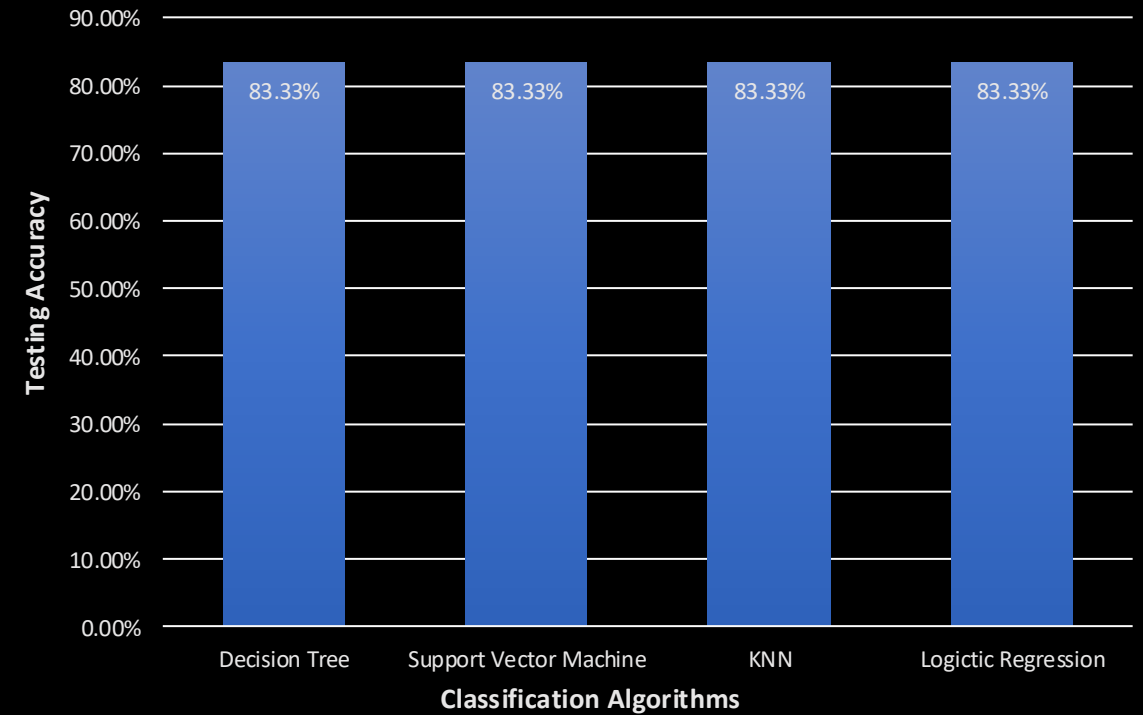
# Section-5

# Classification Accuracy

Training/Validation Accuracy of ML Algorithms



Testing Accuracy of ML Algorithms



# Classification Accuracy

## Key Findings

Four classification models were trained and optimized using **GridSearchCV**.

All models achieved an identical **83.33% test accuracy**, correctly predicting **15 of 18** test samples.

Because the testing dataset is relatively small, the test accuracy alone does not differentiate the models.

Cross-validation accuracy on the training dataset provides a more reliable basis for comparing model performance.

# Predictive Analysis: Best Model Selection

## Best Model: Decision Tree Classifier

- Cross-Validation Accuracy: 86.25%
- Test Accuracy: 83.33%

| Model                  | Cross-Validation Accuracy | Test Accuracy |
|------------------------|---------------------------|---------------|
| <b>Decision Tree</b>   | <b>86.25%</b>             | <b>83.33%</b> |
| Support Vector Machine | 84.82%                    | 83.33%        |
| K-Nearest Neighbors    | 83.39%                    | 83.33%        |
| Logistic Regression    | 82.14%                    | 83.33%        |

# Predictive Analysis: Best Model Selection

## Why Was the Decision Tree Selected?

- All four models achieved the same **83.33% test accuracy**, resulting in a tie.
- To distinguish between the models, **cross-validation accuracy (GridSearchCV best\_score\_)** was used as the secondary evaluation metric.
- The **Decision Tree classifier** achieved the **highest cross-validation accuracy of 86.25%**, indicating the strongest performance during training and validation.
- Based on its superior validation performance while maintaining the same test accuracy as the other models, the **Decision Tree classifier** was selected as the best-performing model.



# Predictive Analysis: Confusion Matrix of the Best Model

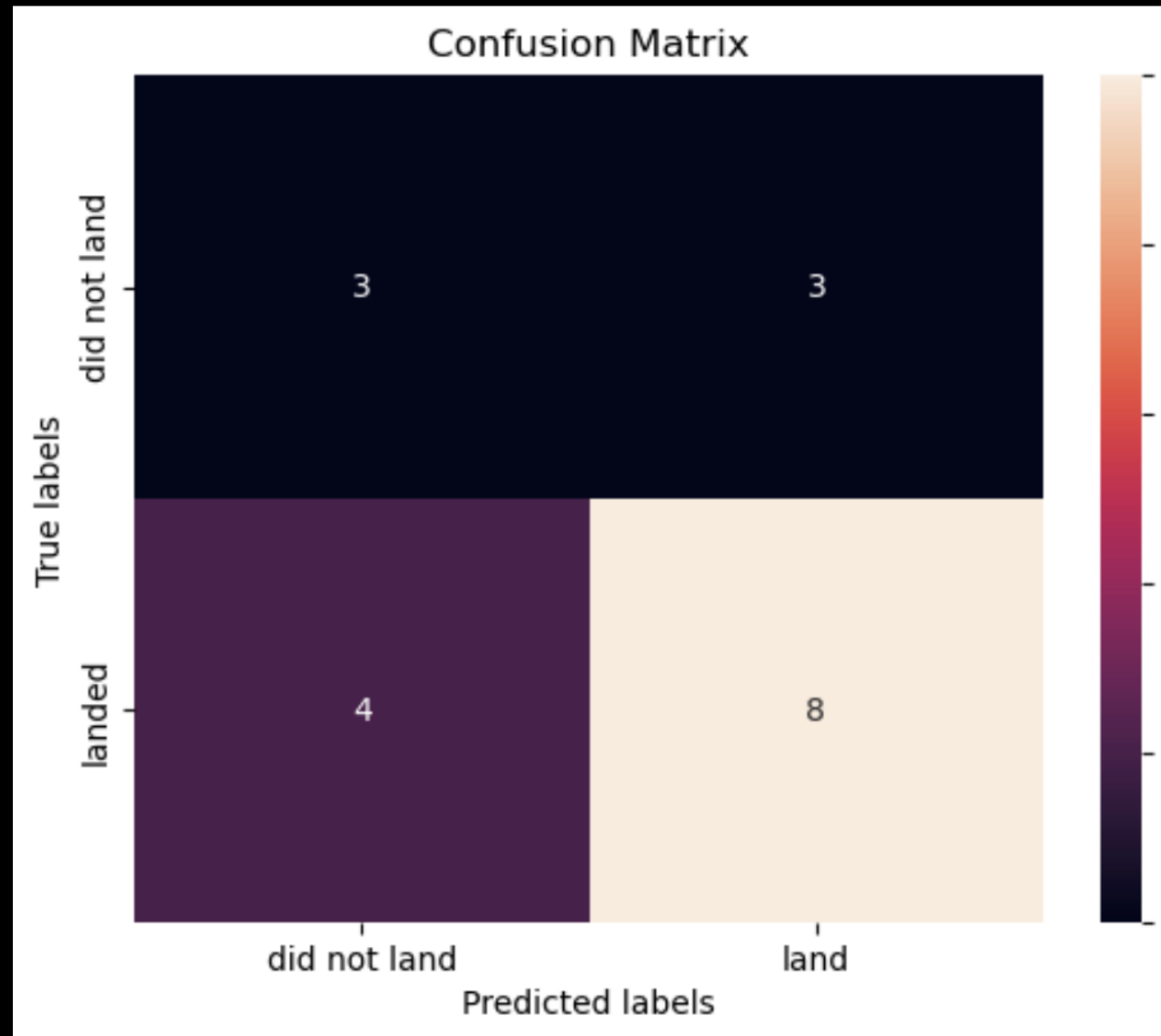
It summarizes:

**True Positives:**  
Successful landings  
correctly predicted.

**True Negatives:**  
Failed landings  
correctly predicted.

**False Positives:**  
Failed landings  
predicted as  
successful.

**False Negatives:**  
Successful landings  
predicted as  
failures.





## Predictive Analysis: Confusion Matrix of the Best Model

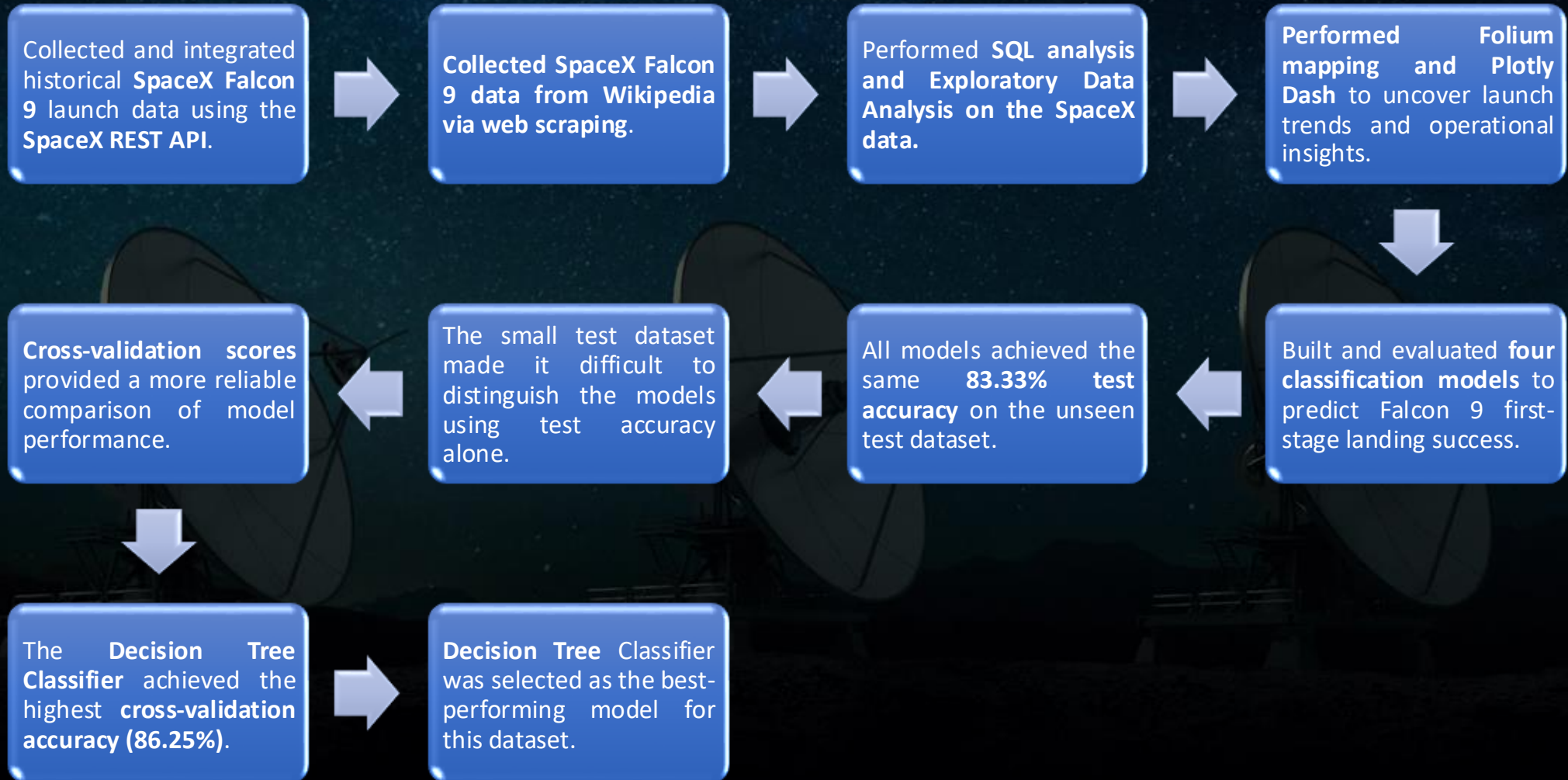
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### Key Findings

- Decision Tree achieved the **highest cross-validation accuracy (86.25%)**.
- Maintained **83.33% accuracy** on unseen test data.
- Correctly classified **15 of 18** launch records.
- Demonstrated reliable predictive performance for Falcon 9 landing success.



# Conclusion



Files

main

Go to file

10\_Applied\_Data\_Science\_Project

1\_Project\_Coding\_Part

Assets

1\_spacex\_api\_scraped.csv

2\_spacex\_web\_scraped.csv

3\_spacex\_api\_scraped\_dataset\_part-1.csv

4\_spacex\_api\_scraped\_dataset\_part-2.csv

5\_spacex\_dataset.csv

6\_my\_data1.db

7\_spacex\_api\_scraped\_dataset\_part-3.csv

8\_spacex\_launch\_geo.csv

9\_spacex\_launch\_dash.csv

1\_Spacex\_Data\_Collection\_Using\_API.ipynb

2\_Spacex\_Data\_Collection\_Webscraping.ipynb

3\_Spacex\_Data\_Wrangling.ipynb

4\_Spacex\_Dataset\_EDA\_Sql\_Sqlite.ipynb

5\_Spacex\_Dataset\_EDA\_Feature\_Engineering.ipynb

6\_Spacex\_Dataset\_Visualizations\_with\_Folium.ipynb

7\_Spacex\_Dataset\_Visualizations\_with\_Plotly\_Dash.py

8\_SpaceX\_Machine\_Learning\_Prediction.ipynb

2\_Project\_Notes\_Materials

1\_Data\_Science\_Capstone\_Project\_Notes.docx

Data\_Science\_Specialization / 10\_Applied\_Data\_Science\_Project / 1\_Pro

KashifMaqbool

Completing Data Science Final Project

| Name                                                | Last commit me |
|-----------------------------------------------------|----------------|
| ..                                                  |                |
| Assets                                              | Completing Da  |
| 1_Spacex_Data_Collection_Using_API.ipynb            | Collecting spa |
| 2_Spacex_Data_Collection_Webscraping.ipynb          | Collecting spa |
| 3_Spacex_Data_Wrangling.ipynb                       | Performing Exp |
| 4_Spacex_Dataset_EDA_Sql_Sqlite.ipynb               | Performing Exp |
| 5_Spacex_Dataset_EDA_Feature_Engineering.ipynb      | Performing Exp |
| 6_Spacex_Dataset_Visualizations_with_Folium.ipynb   | Performing Ge  |
| 7_Spacex_Dataset_Visualizations_with_Plotly_Dash.py | Completing Da  |
| 8_SpaceX_Machine_Learning_Prediction.ipynb          | Completing Da  |

# Appendix

## Project Resources

- GitHub Repository
- Jupyter Notebooks
- SQL Scripts
- Python Source Code
- Dataset
- PowerPoint Presentation

[SpaceX DS capstone project GitHub repository link](#)

**Thank You!**